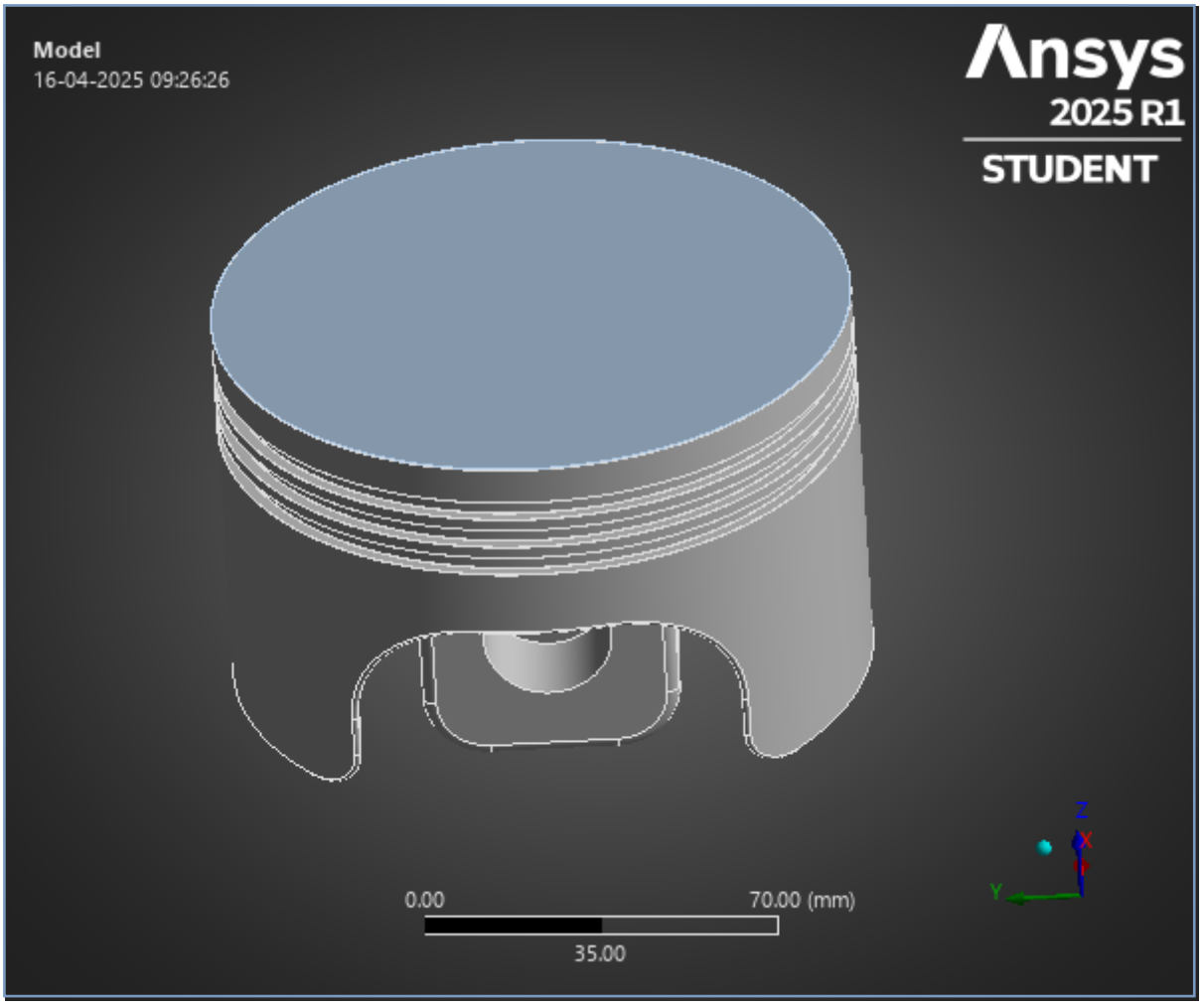
From:
To:
Date: 4/16/2025 3:57:17 AM
Subject:

Project

First Saved	Wednesday, April 16, 2025
Last Saved	Wednesday, April 16, 2025
Product Version	2025 R1
Save Project Before Solution	No
Save Project After Solution	No



Contents

- Units
- Model (A4)
  - Geometry Imports
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  - Aluminum Alloy
  - TBC coating

Units

TABLE 1

Unit System	Metric (mm, kg, N, s, mV, mA) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2
Model (A4) > Geometry Imports

Object Name	*Geometry Imports*
State	Solved

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	*Geometry Import (A3)*
State	Solved
Definition	
Source	D:\HEAT TRANSFER PROECT\transient .2mm_files\dp0\SYS\DM\SYS.dsco
Type	Discovery
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes

Named Selection Key	
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

Object Name	*Geometry*
State	Fully Defined
Definition	
Source	D:\HEAT TRANSFER PROECT\transient .2mm_files\dp0\SYS\DM\SYS.dsco
Type	Discovery
Length Unit	Meters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	127. mm
Length Y	127. mm
Length Z	82.75 mm
Properties	
Volume	4.408e+005 mm³
Mass	1.2263 kg
Scale Factor Value	1.
Statistics	
Bodies	2
Active Bodies	2
Nodes	94705
Elements	55039
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	Yes
Parameters	Independent
Parameter Key	
Attributes	Yes
Attribute Key	
Named Selections	Yes
Named Selection Key	
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	Yes
Coordinate System Key	

Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	*Piston:1\Body1*	*Piston Assembly v14\Body1*
State	Meshed	
Graphics Properties		
Visible	Yes	
Transparency	1	
Definition		
Suppressed	No	
Stiffness Behavior	Flexible	
Coordinate System	Default Coordinate System	
Reference Temperature	By Environment	
Treatment	None	
Material		
Assignment	Aluminum Alloy	TBC coating
Nonlinear Effects	Yes	
Thermal Strain Effects	Yes	
Bounding Box		
Length X	127. mm	126.9 mm
Length Y	127. mm	126.9 mm
Length Z	82.55 mm	0.2 mm
Properties		
Volume	4.3827e+005 mm³	2529.5 mm³
Mass	1.214 kg	1.2268e-002 kg
Centroid X	-8.6651e-004 mm	-2.1106e-007 mm
Centroid Y	-1.4374e-003 mm	4.6102e-015 mm
Centroid Z	-27.291 mm	0.1 mm
Moment of Inertia Ip1	2036. kg·mm²	12.336 kg·mm²
Moment of Inertia Ip2	1846.9 kg·mm²	12.336 kg·mm²
Moment of Inertia Ip3	2569.5 kg·mm²	24.671 kg·mm²
Statistics		
Nodes	89708	4997
Elements	54352	687
Mesh Metric	None	
CAD Attributes		
PartTolerance:	0.00000001	
Color:130.130.130		
Color:175.146.143		

TABLE 6
Model (A4) > Materials

Object Name	*Materials*
State	Fully Defined
Statistics	
Materials	3
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	*Global Coordinate System*
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0.
Origin	
Origin X	0. mm
Origin Y	0. mm
Origin Z	0. mm
Directional Vectors	
X Axis Data	[1. 0. 0.]
Y Axis Data	[0. 1. 0.]
Z Axis Data	[0. 0. 1.]
Transfer Properties	
Source	
Read Only	No

Connections

TABLE 8
Model (A4) > Connections

Object Name	*Connections*
State	Fully Defined
Auto Detection	
Generate Automatic Connection On Refresh	Yes
Transparency	
Enabled	Yes
Statistics	
Contacts	1
Active Contacts	1
Joints	0
Active Joints	0
Beams	0
Active Beams	0
Bearings	0
Active Bearings	0
Springs	0
Active Springs	0
Body Interactions	0
Active Body Interactions	0

TABLE 9
Model (A4) > Connections > Contacts

Object Name	*Contacts*
State	Fully Defined
Definition	
Connection Type	Contact
Scope	
Scoping Method	Geometry Selection
Geometry	All Bodies
Auto Detection	
Tolerance Type	Slider
Tolerance Slider	0.
Tolerance Value	0.49438 mm
Use Range	No
Face/Face	Yes
Face-Face Angle Tolerance	75. °
Face Overlap Tolerance	Off
Cylindrical Faces	Include

Face/Edge	No
Edge/Edge	No
Priority	Include All
Group By	Bodies
Search Across	Bodies
Statistics	
Connections	1
Active Connections	1

TABLE 10
Model (A4) > Connections > Contacts > Contact Regions

Object Name	*Contact Region*
State	Fully Defined
Scope	
Scoping Method	Geometry Selection
Contact	2 Faces
Target	2 Faces
Contact Bodies	Piston:1\Body1
Target Bodies	Piston Assembly v14\Body1
Protected	No
Definition	
Type	Bonded
Scope Mode	Automatic
Behavior	Program Controlled
Trim Contact	Program Controlled
Trim Tolerance	0.49438 mm
Contact APDL Name	
Target APDL Name	
Suppressed	No
Display	
Element Normals	No
Advanced	
Formulation	Program Controlled
Small Sliding	Program Controlled
Detection Method	Program Controlled
Elastic Slip Tolerance	Program Controlled
Thermal Conductance	Program Controlled
Pinball Region	Program Controlled
Geometric Modification	
Contact Geometry Correction	None
Target Geometry Correction	None

Mesh

TABLE 11
Model (A4) > Mesh

Object Name	*Mesh*
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	5.0 mm
Sizing	
Use Adaptive Sizing	No
Growth Rate	Default (1.85)
Max Size	Default (10.0 mm)
Mesh Defeaturing	Yes
Defeature Size	Default (2.5e-002 mm)
Capture Curvature	No

Capture Proximity	No
Bounding Box Diagonal	197.75 mm
Average Surface Area	876.49 mm²
Minimum Edge Length	0.63636 mm
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (5.e-002)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0.272
Maximum Layers	5
Growth Rate	1.2
Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Default (4.5e-002 mm)
Generate Pinch on Refresh	No
Automatic Methods	
Sheet Body Method	Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	94705
Elements	55039
Show Detailed Statistics	No

Transient Thermal (A5)

TABLE 12
Model (A4) > Analysis

Object Name	*Transient Thermal (A5)*
State	Solved
Definition	
Physics Type	Thermal
Analysis Type	Transient
Solver Target	Mechanical APDL
Options	
Generate Input Only	No

TABLE 13
Model (A4) > Transient Thermal (A5) > Initial Condition

Object Name	*Initial Temperature*
State	Fully Defined
Definition	
Initial Temperature	Uniform Temperature
Initial Temperature Value	22. °C

TABLE 14
Model (A4) > Transient Thermal (A5) > Analysis Settings

Object Name	*Analysis Settings*
State	Fully Defined
Step Controls	

Number Of Steps	71.
Current Step Number	71.
Step End Time	71.1 s
Auto Time Stepping	Program Controlled
Initial Time Step	1.0014e-002 s
Minimum Time Step	1.0014e-003 s
Maximum Time Step	0.10014 s
Time Integration	On
Solver Controls	
Solver Type	Program Controlled
Radiosity Controls	
Radiosity Solver	Program Controlled
Flux Convergence	1.e-004
Maximum Iteration	1000.
Solver Tolerance	1.e-007 W/mm²
Over Relaxation	0.1
Hemicube Resolution	10.
Nonlinear Controls	
Heat Convergence	Program Controlled
Temperature Convergence	Program Controlled
Line Search	Program Controlled
Nonlinear Formulation	Program Controlled
Advanced	
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Calculate Thermal Flux	Yes
Contact Data	Yes
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	D:\HEAT TRANSFER PROECT\transient .2mm_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	Yes
Solver Units	Active System
Solver Unit System	nmm

TABLE 15
Model (A4) > Transient Thermal (A5) > Analysis Settings
Step-Specific "Step Controls"

Step	Step End Time	Initial Time Step	Minimum Time Step	Maximum Time Step
1	1.0014 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
2	2.0028 s			
3	3.0042 s			
4	4.0056 s			
5	5.007 s			
6	6.0085 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
7	7.0099 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
8	8.0113 s			
9	9.0127 s			
10	10.014 s			
11	11.015 s			

12	12.017 s			
13	13.018 s			
14	14.02 s			
15	15.021 s			
16	16.023 s			
17	17.024 s			
18	18.025 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
19	19.027 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
20	20.028 s			
21	21.03 s			
22	22.031 s			
23	23.032 s			
24	24.034 s			
25	25.035 s			
26	26.037 s			
27	27.038 s			
28	28.039 s			
29	29.041 s			
30	30.042 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
31	31.044 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
32	32.045 s			
33	33.047 s			
34	34.048 s			
35	35.049 s			
36	36.051 s			
37	37.052 s			
38	38.053 s			
39	39.055 s			
40	40.056 s			
41	41.058 s			
42	42.059 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
43	43.061 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
44	44.062 s			
45	45.063 s			
46	46.065 s			
47	47.066 s			
48	48.068 s			
49	49.069 s			
50	50.07 s			
51	51.072 s			
52	52.073 s			
53	53.075 s			
54	54.076 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
55	55.078 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
56	56.079 s			
57	57.08 s			
58	58.082 s			
59	59.083 s			
60	60.084 s			
61	61.086 s			
62	62.087 s			
63	63.089 s			
64	64.09 s			
65	65.091 s			
66	66.093 s	1.0015e-002 s	1.0015e-003 s	0.10015 s
67	67.094 s	1.0014e-002 s	1.0014e-003 s	0.10014 s
68	68.096 s			
69	69.097 s			
70	70.099 s			
71	71.1 s			

TABLE 16
Model (A4) > Transient Thermal (A5) > Loads

Object Name	*Temperature*	*Convection*
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	1 Body	
Apply To	Exterior Faces Only	
Definition		
Type	Temperature	Convection
Magnitude	Tabular Data	
Suppressed	No	
Film Coefficient		Tabular Data
Coefficient Type		Difference of Surface and Bulk Temp
Ambient Temperature		22. °C (step applied)
Convection Matrix		Program Controlled
Element APDL Name		
Edit Data For		Film Coefficient
Tabular Data		
Independent Variable	Time	Temperature
Graph Controls		
X-Axis		

FIGURE 1
Model (A4) > Transient Thermal (A5) > Temperature

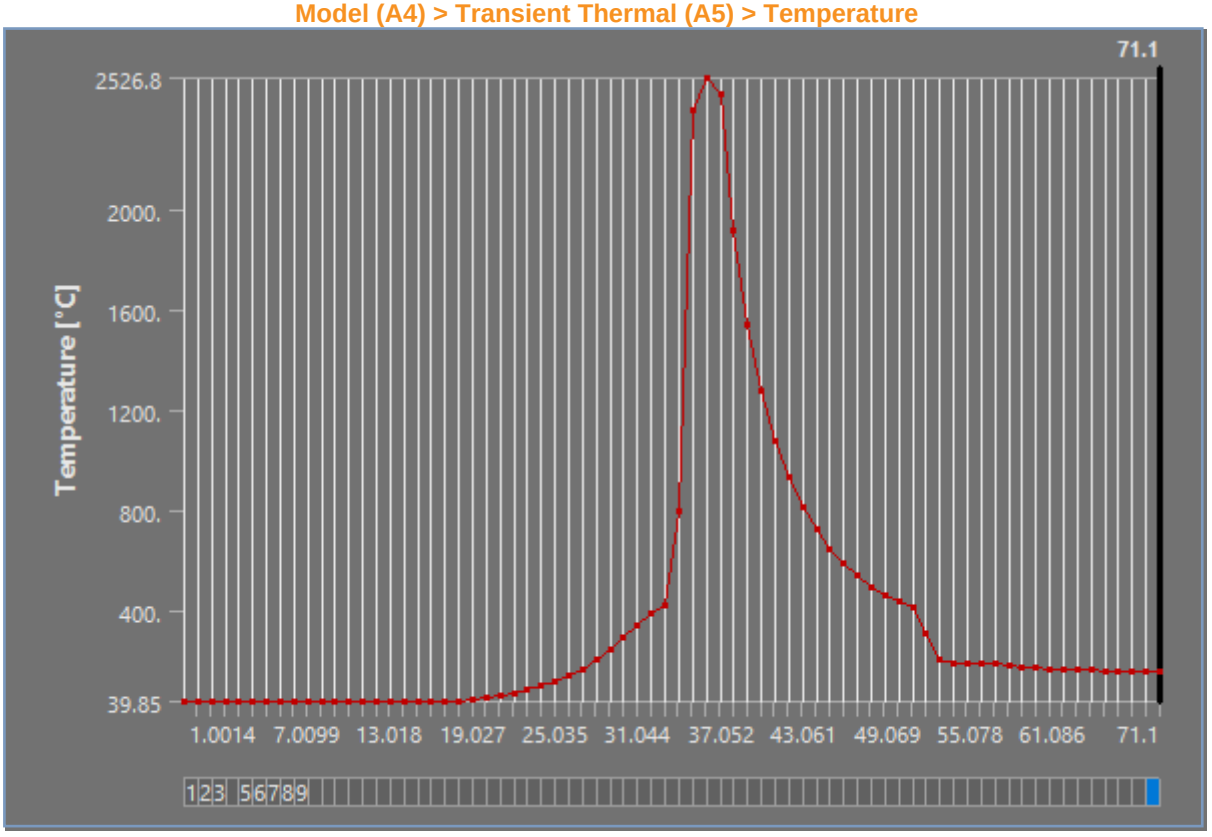


TABLE 17
Model (A4) > Transient Thermal (A5) > Temperature

Steps	Time [s]	Temperature [°C]
1	0.	39.85
	1.0014	
2	2.0028	
3	3.0042	
4	4.0056	
5	5.007	
6	6.0085	
7	7.0099	

8	8.0113	
9	9.0127	
10	10.014	
11	11.015	
12	12.017	
13	13.018	
14	14.02	
15	15.021	
16	16.023	
17	17.024	
18	18.025	39.899
19	19.027	41.062
20	20.028	43.803
21	21.03	48.196
22	22.031	54.358
23	23.032	62.462
24	24.034	72.744
25	25.035	85.52
26	26.037	101.2
27	27.038	120.31
28	28.039	143.53
29	29.041	171.67
30	30.042	205.63
31	31.044	246.19
32	32.045	293.32
33	33.047	344.6
34	34.048	392.55
35	35.049	423.26
36	36.051	799.
37	37.052	2397.4
38	38.053	2526.8
39	39.055	2463.5
40	40.056	1918.6
41	41.058	1546.3
42	42.059	1280.3
43	43.061	1083.4
44	44.062	933.49
45	45.063	816.81
46	46.065	724.31
47	47.066	649.92
48	48.068	589.46
49	49.069	539.92
50	50.07	499.14
51	51.072	465.53
52	52.073	437.91
53	53.075	415.37
54	54.076	313.04
55	55.078	210.56
56	56.079	190.14
57	57.08	
58	58.082	189.85
59	59.083	
60	60.084	184.81
61	61.086	179.47
62	62.087	174.87
63	63.089	171.04
64	64.09	167.98
65	65.091	165.6
66	66.093	163.81
67	67.094	162.5
68	68.096	161.58

69	69.097	160.95
70	70.099	160.53
71	71.1	160.26

FIGURE 2
Model (A4) > Transient Thermal (A5) > Convection

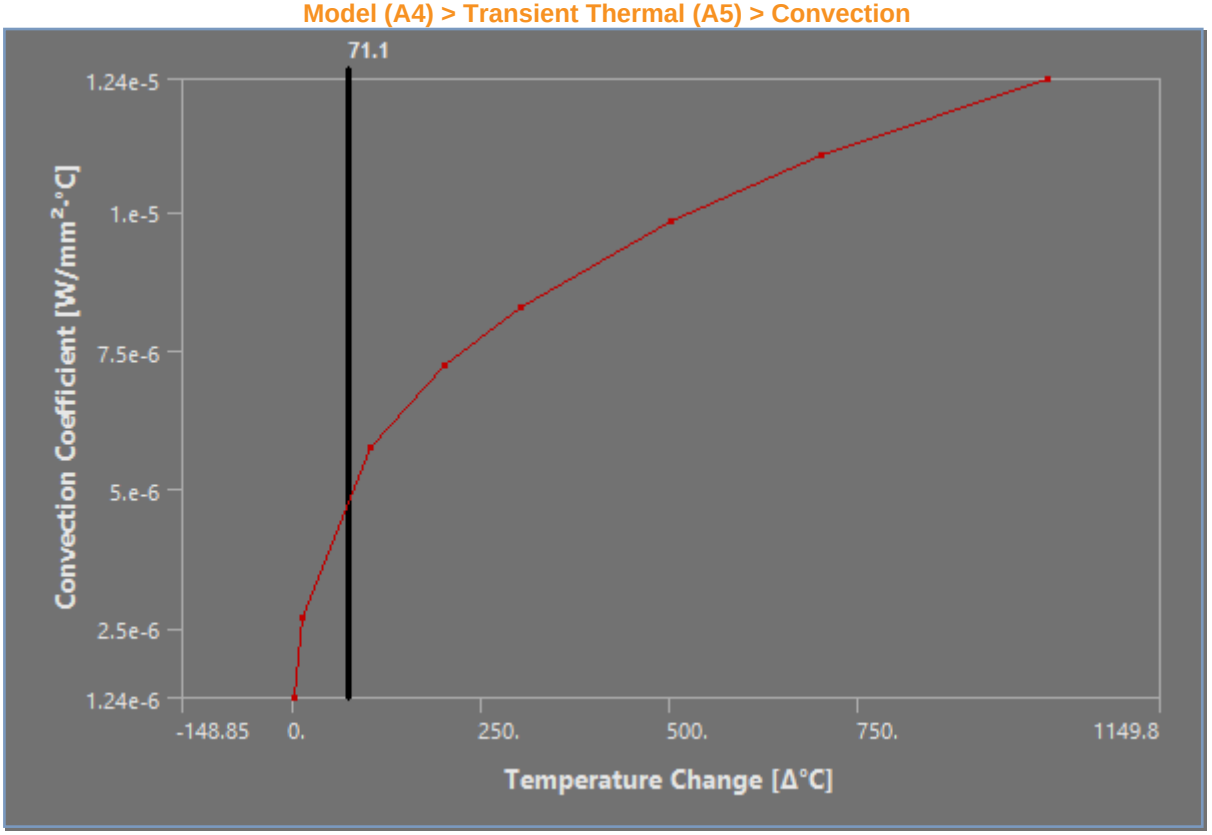


TABLE 18
Model (A4) > Transient Thermal (A5) > Convection

Temperature Change [?°C]	Convection Coefficient [W/mm²·°C]
1.	1.24e-006
10.	2.67e-006
100.	5.76e-006
200.	7.25e-006
300.	8.3e-006
500.	9.84e-006
700.	1.101e-005
1000.	1.24e-005

Solution (A6)

TABLE 19
Model (A4) > Transient Thermal (A5) > Solution

Object Name	*Solution (A6)*
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1.
Refinement Depth	2.
Information	
Status	Done
MAPDL Elapsed Time	21 m 28 s
MAPDL Memory Used	1.043 GB
MAPDL Result File Size	12.426 GB
Post Processing	
Beam Section Results	No

TABLE 20
Model (A4) > Transient Thermal (A5) > Solution (A6) > Solution Information

Object Name	*Solution Information*
State	Solved
Solution Information	
Solution Output	Solver Output
Update Interval	2.5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 21
Model (A4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Result Charts

Object Name	*Temperature - Global Maximum*	*Temperature - Global Minimum*
State	Solved	
Scope		
Scoping Method	Global Maximum	Global Minimum
Definition		
Type	Temperature	
Suppressed	No	
Results		
Minimum	39.85 °C	16.991 °C
Maximum	2526.8 °C	421.23 °C

FIGURE 3
Model (A4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Temperature - Global Maximum

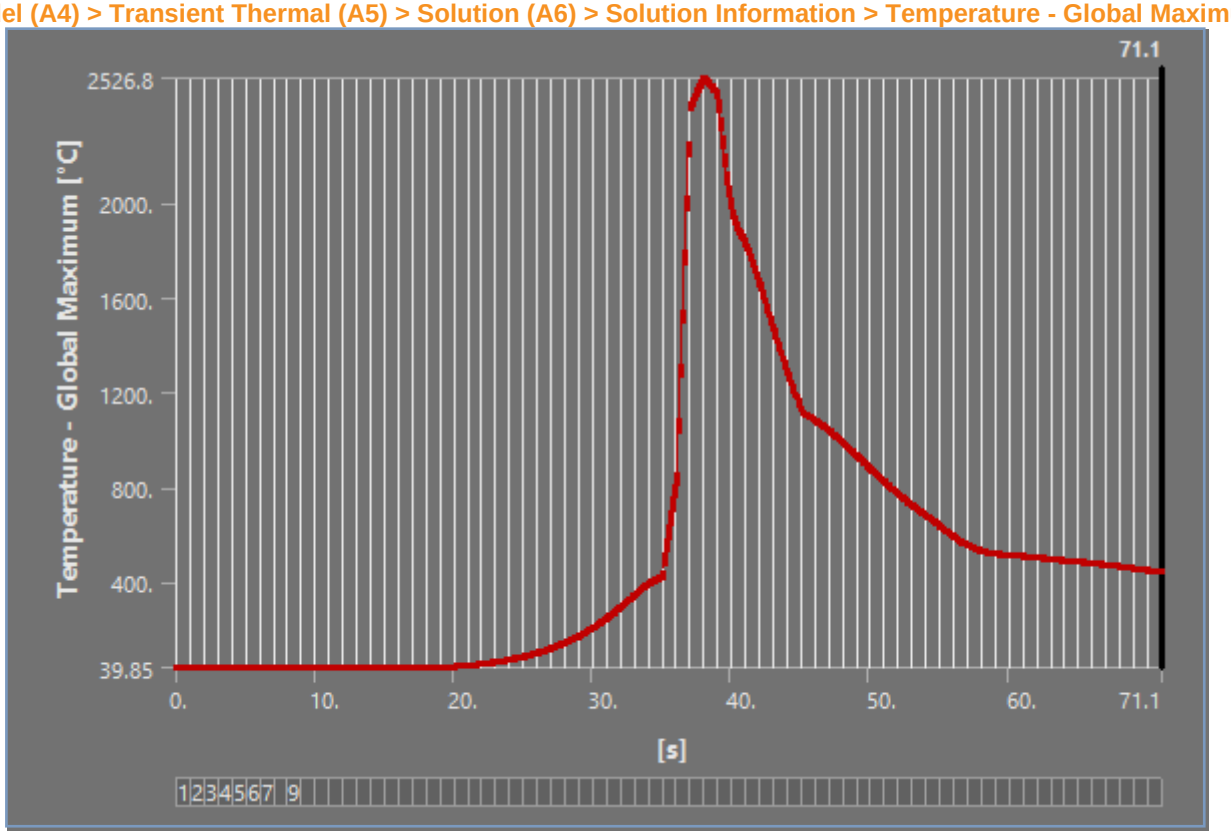


FIGURE 4
Model (A4) > Transient Thermal (A5) > Solution (A6) > Solution Information > Temperature - Global Minimum

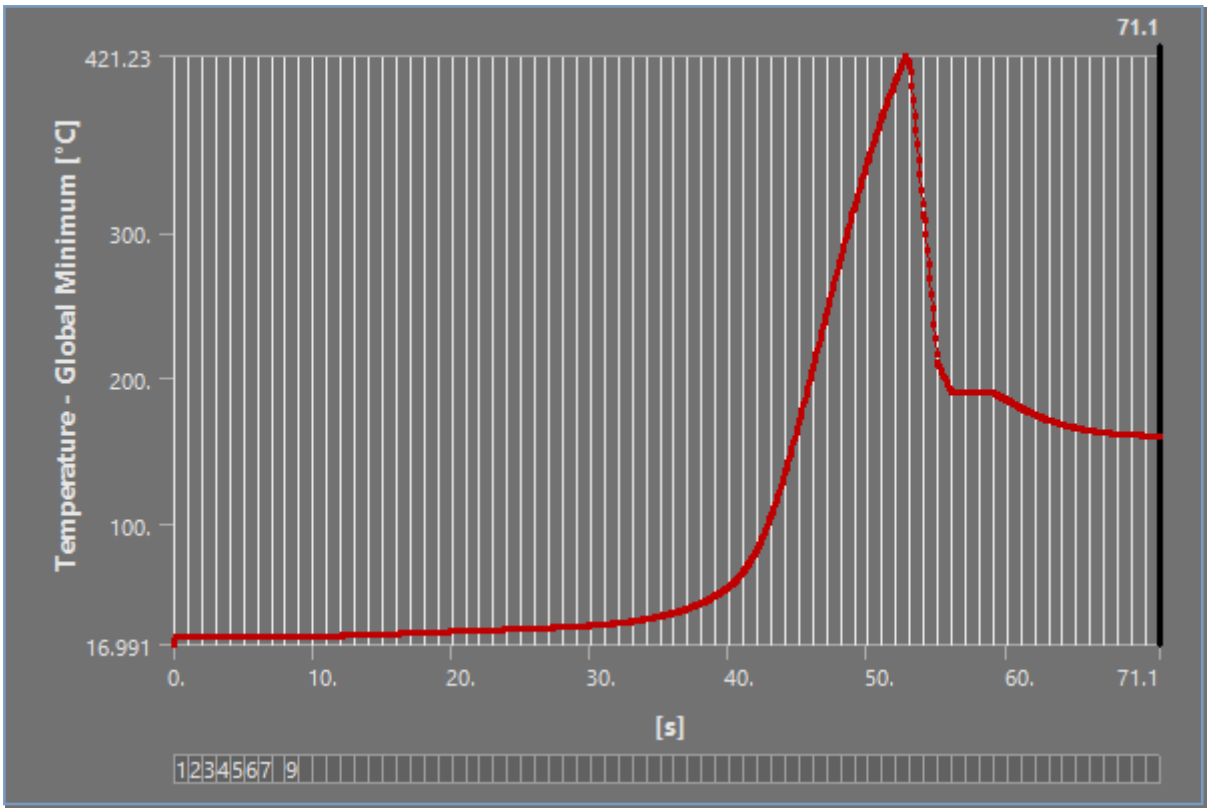


TABLE 22
Model (A4) > Transient Thermal (A5) > Solution (A6) > Results

Object Name	*Temperature*	*Total Heat Flux*
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Temperature	Total Heat Flux
By	Time	
Display Time	37.95 s	71.1 s
Separate Data by Entity	No	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Results		
Minimum	45.271 °C	2.5656e-014 W/mm²
Maximum	2513.5 °C	3.9665 W/mm²
Average	1036.3 °C	0.47754 W/mm²
Minimum Occurs On	Piston:1\Body1	Piston Assembly v14\Body1
Maximum Occurs On	Piston Assembly v14\Body1	Piston:1\Body1
Minimum Value Over Time		
Minimum	16.991 °C	1.6714e-017 W/mm²
Maximum	421.23 °C	2.5592e-008 W/mm²
Maximum Value Over Time		
Minimum	39.85 °C	0.17387 W/mm²
Maximum	2526.8 °C	51.405 W/mm²
Information		
Time	37.95 s	71.1 s
Load Step	38	71
Substep	36	13
Iteration Number	601	1079
Integration Point Results		
Display Option		Averaged
Average Across Bodies		No

FIGURE 5
Model (A4) > Transient Thermal (A5) > Solution (A6) > Temperature

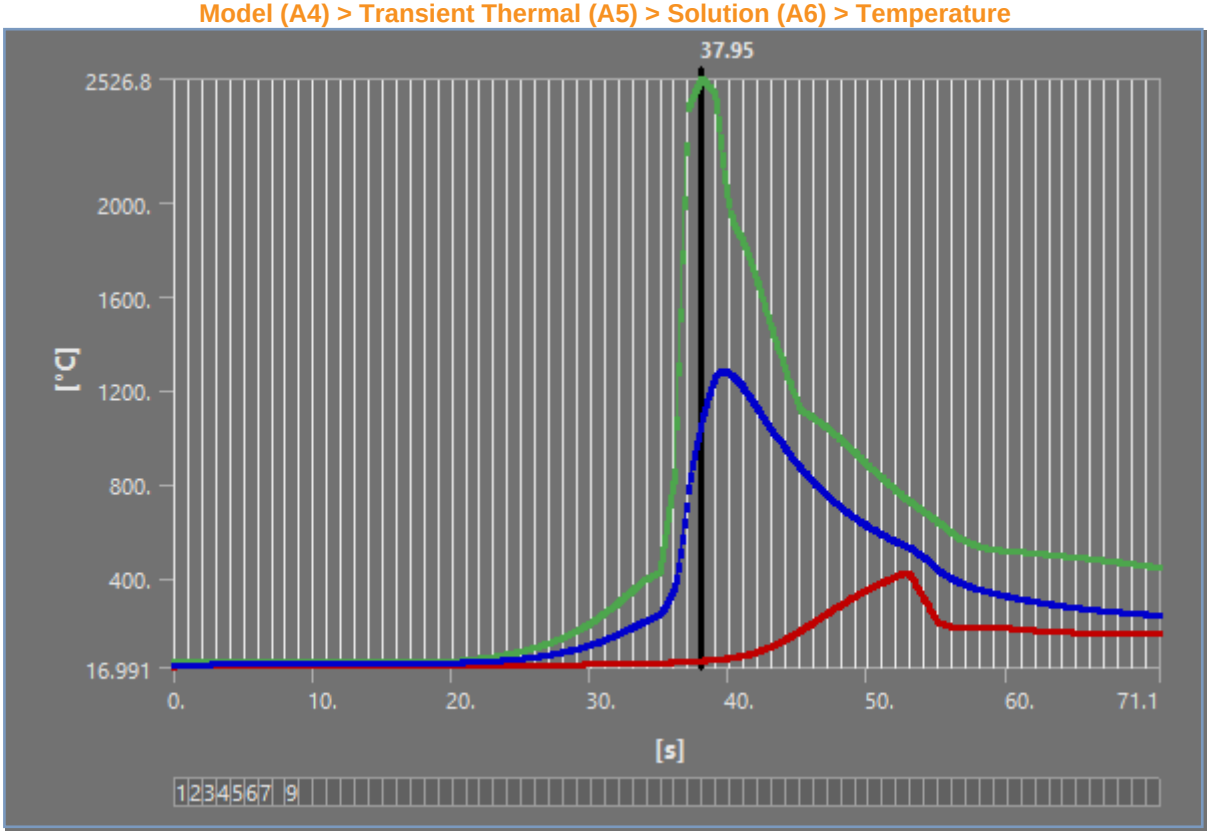


TABLE 23
Model (A4) > Transient Thermal (A5) > Solution (A6) > Temperature

Time [s]	Minimum [°C]	Maximum [°C]	Average [°C]
1.0014e-002	16.991	39.995	23.648
1.7163e-002	18.878	39.93	23.766
2.1855e-002	19.777	39.863	23.83
2.6548e-002	20.466	39.85	23.879
3.7797e-002	21.311		23.959
5.6425e-002	21.741		24.063
8.5663e-002	22.	39.872	24.206
0.13041		39.88	24.406
0.19908		39.871	24.679
0.29922		39.859	25.031
0.39936		39.856	25.35
0.4995		39.854	25.646
0.59964		39.853	25.923
0.69978		39.852	26.186
0.79992		39.851	26.437
0.90006		39.85	26.678
1.0014			26.911
1.0114			26.934
1.0214			26.958
1.0515			27.026
1.1416			27.226
1.2417			27.439
1.3419			27.645
1.442			27.843
1.5422			28.034
1.6423			28.219
1.7424			28.397
1.8426			28.57
1.9427			28.736
2.0028			28.834
2.0128			28.85

2.0228			28.867
2.0529			28.915
2.143			29.055
2.2431			29.206
2.3433			29.352
2.4434			29.494
2.5436			29.631
2.6437			29.763
2.7438			29.891
2.844			30.016
2.9441	22.001		30.136
3.0042			30.207
3.0142			30.219
3.0242			30.231
3.0543			30.266
3.1444			30.368
3.2445			30.479
3.3447	22.002		30.586
3.4448			30.69
3.545			30.791
3.6451	22.003		30.889
3.7452	22.004		30.984
3.8454			31.077
3.9455	22.005		31.167
4.0056	22.006		31.221
4.0156			31.23
4.0256			31.239
4.0557			31.265
4.1458	22.007		31.342
4.2459	22.009		31.425
4.3461	22.01		31.507
4.4462	22.012		31.586
4.5464	22.013		31.663
4.6465	22.015		31.739
4.7466	22.018		31.813
4.8468	22.02		31.884
4.9469	22.023		31.955
5.007	22.024		31.996
5.017	22.025		32.003
5.027			32.01
5.0571	22.026		32.03
5.1472	22.029		32.091
5.2474	22.032		32.156
5.3475	22.036		32.22
5.4477	22.04		32.283
5.5478	22.044		32.344
5.648	22.049		32.404
5.7481	22.054		32.463
5.8483	22.059		32.521
5.9484	22.065		32.577
6.0085	22.068		32.611
6.0185	22.069		32.616
6.0285			32.622
6.0586	22.071		32.638
6.1487	22.077		32.687
6.2488	22.084		32.74
6.349	22.091		32.792
6.4491	22.098		32.843
6.5493	22.106		32.894
6.6494	22.114		32.943
6.7495	22.123		32.991

6.8497	22.132		33.039
6.9498	22.142		33.085
7.0099	22.148		33.113
7.0199	22.149		33.118
7.0299	22.15		33.122
7.06	22.153		33.136
7.1501	22.162		33.176
7.2502	22.173		33.221
7.3504	22.184		33.264
7.4505	22.195		33.307
7.5507	22.207		33.349
7.6508	22.22		33.391
7.7509	22.233		33.432
7.8511	22.246		33.472
7.9512	22.26		33.511
8.0113	22.268		33.535
8.0213	22.27		33.539
8.0313	22.271		33.543
8.0614	22.275		33.554
8.1515	22.289		33.589
8.2516	22.304		33.627
8.3518	22.319		33.664
8.4519	22.335		33.7
8.5521	22.352		33.737
8.6522	22.368		33.772
8.7523	22.386		33.807
8.8525	22.403		33.842
8.9526	22.421		33.876
9.0127	22.432		33.897
9.0227	22.434		33.9
9.0327	22.436		33.903
9.0628	22.442		33.913
9.1529	22.459		33.943
9.253	22.478		33.976
9.3532	22.498		34.009
9.4533	22.518		34.041
9.5535	22.538		34.072
9.6536	22.559		34.103
9.7537	22.581		34.134
9.8539	22.602		34.165
9.954	22.624		34.195
10.014	22.638		34.213
10.024	22.64		34.216
10.034	22.642		34.219
10.064	22.649		34.228
10.154	22.67		34.254
10.254	22.693		34.283
10.355	22.717		34.312
10.455	22.74		34.34
10.555	22.765		34.368
10.655	22.789		34.396
10.755	22.814		34.423
10.855	22.84		34.45
10.955	22.866		34.477
11.016	22.881		34.493
11.026	22.884		34.495
11.036	22.886		34.498
11.066	22.894		34.506
11.156	22.918		34.53
11.256	22.944		34.555
11.356	22.971		34.581

11.456	22.999		34.606
11.556	23.026		34.632
11.656	23.054		34.656
11.757	23.082		34.681
11.857	23.111		34.705
11.957	23.139		34.73
12.017	23.157		34.744
12.027	23.16		34.746
12.037	23.163		34.749
12.067	23.171		34.756
12.157	23.198		34.777
12.257	23.227		34.801
12.357	23.257		34.824
12.458	23.287		34.847
12.558	23.317		34.869
12.658	23.348		34.892
12.758	23.379		34.914
12.858	23.41		34.936
12.958	23.441		34.958
13.018	23.46		34.971
13.028	23.463		34.974
13.038	23.466		34.976
13.068	23.475		34.982
13.158	23.504		35.002
13.259	23.536		35.023
13.359	23.568		35.044
13.459	23.6		35.065
13.559	23.633		35.086
13.659	23.665		35.107
13.759	23.698		35.127
13.859	23.731		35.147
13.96	23.765		35.167
14.02	23.785		35.179
14.03	23.788		35.181
14.04	23.791		35.183
14.07	23.801		35.189
14.16	23.832		35.207
14.26	23.865		35.227
14.36	23.899		35.246
14.46	23.933		35.265
14.56	23.967		35.284
14.661	24.002		35.303
14.761	24.036		35.322
14.861	24.071		35.341
14.961	24.106		35.359
15.021	24.126		35.37
15.031	24.13		35.372
15.041	24.133		35.374
15.071	24.144		35.38
15.161	24.175		35.396
15.261	24.211		35.414
15.362	24.246		35.432
15.462	24.281		35.45
15.562	24.317		35.468
15.662	24.352		35.485
15.762	24.388		35.503
15.862	24.424		35.52
15.962	24.459		35.537
16.022	24.481		35.547
16.033	24.484		35.549
16.043	24.488		35.551

16.073	24.499		35.556
16.163	24.531		35.571
16.263	24.567		35.588
16.363	24.604		35.605
16.463	24.64		35.621
16.563	24.676		35.638
16.663	24.713		35.654
16.764	24.749		35.67
16.864	24.786		35.687
16.964	24.822		35.703
17.024	24.844		35.712
17.034	24.848		35.714
17.044	24.851	39.851	35.715
17.074	24.862	39.852	35.72
17.164	24.895	39.857	35.735
17.264	24.932	39.862	35.752
17.364	24.969	39.867	35.768
17.465	25.006	39.872	35.785
17.565	25.043	39.877	35.801
17.665	25.08	39.882	35.818
17.765	25.117	39.886	35.834
17.865	25.154	39.891	35.85
17.965	25.191	39.896	35.867
18.025	25.213	39.899	35.877
18.035	25.216	39.911	35.879
18.045	25.22	39.923	35.882
18.075	25.231	39.957	35.891
18.166	25.265	40.062	35.919
18.266	25.302	40.178	35.953
18.366	25.339	40.295	35.989
18.466	25.376	40.411	36.027
18.566	25.413	40.527	36.067
18.666	25.45	40.643	36.108
18.766	25.487	40.76	36.151
18.867	25.525	40.876	36.196
18.967	25.562	40.992	36.242
19.027	25.584	41.062	36.27
19.037	25.588	41.089	36.276
19.047	25.592	41.117	36.283
19.077	25.603	41.199	36.302
19.167	25.636	41.446	36.366
19.267	25.673	41.72	36.441
19.367	25.711	41.994	36.52
19.467	25.748	42.268	36.604
19.568	25.785	42.542	36.691
19.668	25.822	42.816	36.782
19.768	25.859	43.091	36.876
19.868	25.896	43.365	36.974
19.968	25.934	43.639	37.074
20.028	25.956	43.803	37.136
20.038	25.959	43.847	37.147
20.048	25.963	43.891	37.159
20.078	25.974	44.023	37.196
20.168	26.008	44.418	37.313
20.269	26.045	44.858	37.448
20.369	26.082	45.297	37.59
20.469	26.119	45.736	37.737
20.569	26.156	46.175	37.889
20.669	26.193	46.615	38.047
20.769	26.23	47.054	38.209
20.869	26.267	47.493	38.375

20.97	26.304	47.932	38.547
21.03	26.326	48.196	38.651
21.04	26.33	48.258	38.67
21.05	26.333	48.319	38.689
21.08	26.344	48.504	38.748
21.17	26.378	49.059	38.932
21.27	26.414	49.675	39.144
21.37	26.451	50.291	39.363
21.47	26.488	50.907	39.589
21.57	26.525	51.524	39.822
21.67	26.562	52.14	40.061
21.771	26.598	52.756	40.306
21.871	26.635	53.372	40.556
21.971	26.671	53.988	40.812
22.031	26.693	54.358	40.967
22.041	26.697	54.439	40.995
22.051	26.701	54.52	41.024
22.081	26.712	54.763	41.109
22.171	26.745	55.493	41.374
22.271	26.781	56.303	41.677
22.371	26.818	57.113	41.99
22.472	26.854	57.924	42.31
22.572	26.89	58.734	42.638
22.672	26.927	59.545	42.973
22.772	26.963	60.355	43.315
22.872	27.	61.165	43.664
22.972	27.036	61.976	44.02
23.032	27.058	62.462	44.235
23.042	27.061	62.565	44.274
23.052	27.065	62.668	44.312
23.082	27.076	62.976	44.429
23.173	27.108	63.901	44.79
23.273	27.145	64.93	45.201
23.373	27.181	65.958	45.622
23.473	27.217	66.986	46.052
23.573	27.253	68.014	46.492
23.673	27.29	69.042	46.94
23.773	27.326	70.071	47.396
23.874	27.362	71.099	47.86
23.974	27.399	72.127	48.332
24.034	27.42	72.744	48.618
24.044	27.424	72.872	48.669
24.054	27.428	72.999	48.72
24.084	27.438	73.383	48.873
24.174	27.471	74.532	49.346
24.274	27.508	75.81	49.883
24.374	27.544	77.088	50.432
24.474	27.58	78.365	50.992
24.575	27.617	79.643	51.563
24.675	27.654	80.92	52.143
24.775	27.69	82.198	52.733
24.875	27.727	83.476	53.332
24.975	27.764	84.753	53.94
25.035	27.786	85.52	54.308
25.045	27.79	85.676	54.373
25.055	27.793	85.833	54.438
25.085	27.804	86.304	54.635
25.175	27.838	87.715	55.24
25.276	27.875	89.283	55.927
25.376	27.912	90.851	56.626
25.476	27.95	92.419	57.339

25.576	27.987	93.986	58.064
25.676	28.025	95.554	58.8
25.776	28.063	97.122	59.548
25.876	28.101	98.69	60.306
25.977	28.14	100.26	61.075
26.037	28.163	101.2	61.54
26.047	28.167	101.39	61.622
26.057	28.171	101.58	61.704
26.087	28.182	102.15	61.953
26.177	28.217	103.87	62.715
26.277	28.256	105.79	63.579
26.377	28.296	107.7	64.458
26.477	28.336	109.61	65.353
26.577	28.376	111.52	66.262
26.677	28.416	113.43	67.184
26.778	28.457	115.34	68.119
26.878	28.498	117.25	69.068
26.978	28.539	119.17	70.028
27.038	28.564	120.31	70.615
27.048	28.568	120.54	70.717
27.058	28.573	120.78	70.82
27.088	28.585	121.47	71.13
27.178	28.623	123.56	72.077
27.278	28.666	125.88	73.151
27.378	28.709	128.21	74.244
27.479	28.753	130.53	75.354
27.579	28.797	132.85	76.481
27.679	28.842	135.17	77.625
27.779	28.887	137.49	78.784
27.879	28.933	139.81	79.968
27.979	28.98	142.14	81.167
28.039	29.008	143.53	81.891
28.049	29.013	143.81	82.017
28.059	29.018	144.09	82.143
28.089	29.032	144.94	82.526
28.18	29.075	147.47	83.694
28.28	29.123	150.28	85.017
28.38	29.173	153.1	86.363
28.48	29.223	155.91	87.73
28.58	29.274	158.72	89.117
28.68	29.325	161.54	90.523
28.78	29.378	164.35	91.96
28.881	29.431	167.16	93.413
28.981	29.486	169.98	94.884
29.041	29.519	171.67	95.773
29.051	29.524	172.01	95.927
29.061	29.53	172.35	96.082
29.091	29.546	173.36	96.551
29.181	29.597	176.42	97.984
29.281	29.654	179.82	99.604
29.381	29.713	183.21	101.25
29.481	29.772	186.61	102.93
29.582	29.833	190.01	104.62
29.682	29.894	193.4	106.36
29.782	29.957	196.8	108.11
29.882	30.022	200.2	109.89
29.982	30.087	203.59	111.68
30.042	30.127	205.63	112.77
30.052	30.134	206.04	112.95
30.062	30.14	206.44	113.14
30.092	30.16	207.66	113.71

30.182	30.222	211.31	115.46
30.283	30.291	215.36	117.43
30.383	30.362	219.42	119.44
30.483	30.435	223.48	121.47
30.583	30.509	227.53	123.54
30.683	30.585	231.59	125.64
30.783	30.662	235.64	127.76
30.883	30.741	239.7	129.92
30.984	30.822	243.76	132.09
31.044	30.872	246.19	133.41
31.054	30.88	246.66	133.64
31.064	30.888	247.13	133.86
31.094	30.913	248.55	134.55
31.184	30.989	252.79	136.64
31.284	31.075	257.5	139.
31.384	31.164	262.21	141.39
31.484	31.254	266.93	143.82
31.584	31.347	271.64	146.29
31.685	31.442	276.35	148.78
31.785	31.539	281.07	151.3
31.885	31.638	285.78	153.86
31.985	31.739	290.49	156.44
32.045	31.801	293.32	158.
32.055	31.812	293.83	158.26
32.065	31.822	294.34	158.53
32.095	31.853	295.88	159.33
32.185	31.949	300.5	161.75
32.285	32.058	305.63	164.48
32.386	32.17	310.76	167.24
32.486	32.284	315.88	170.03
32.586	32.401	321.01	172.85
32.686	32.521	326.14	175.71
32.786	32.644	331.27	178.58
32.886	32.769	336.4	181.49
32.986	32.898	341.53	184.42
33.047	32.977	344.6	186.19
33.057	32.99	345.08	186.48
33.067	33.003	345.56	186.77
33.097	33.043	347.	187.65
33.187	33.164	351.32	190.29
33.287	33.303	356.11	193.24
33.387	33.445	360.91	196.2
33.487	33.59	365.7	199.18
33.587	33.739	370.5	202.17
33.687	33.892	375.29	205.18
33.788	34.049	380.08	208.2
33.888	34.209	384.88	211.24
33.988	34.373	389.67	214.29
34.048	34.473	392.55	216.13
34.058	34.49	392.86	216.41
34.068	34.507	393.16	216.7
34.098	34.558	394.09	217.56
34.188	34.713	396.85	220.11
34.288	34.89	399.92	222.92
34.388	35.071	402.99	225.7
34.489	35.256	406.06	228.46
34.589	35.447	409.13	231.2
34.689	35.642	412.21	233.93
34.789	35.841	415.28	236.63
34.889	36.046	418.35	239.33
34.989	36.256	421.42	242.01

35.049	36.384	423.26	243.61
35.059	36.405	427.02	244.2
35.069	36.426	430.78	244.82
35.095	36.482	440.48	246.52
35.146	36.593	459.63	250.16
35.197	36.705	478.75	254.04
35.248	36.819	497.87	258.14
35.299	36.934	516.99	262.45
35.35	37.051	536.11	266.95
35.401	37.169	555.22	271.63
35.452	37.288	574.34	276.48
35.503	37.409	593.46	281.5
35.554	37.531	612.58	286.68
35.605	37.655	631.7	292.01
35.656	37.781	650.82	297.5
35.707	37.908	669.94	303.13
35.758	38.036	689.06	308.9
35.809	38.166	708.18	314.82
35.86	38.298	727.3	320.87
35.911	38.431	746.42	327.05
35.962	38.566	765.53	333.36
36.012	38.703	784.65	339.8
36.051	38.806	799.	344.7
36.061	38.834	814.98	347.12
36.071	38.861	830.96	349.67
36.083	38.893	850.08	352.81
36.095	38.926	869.2	356.04
36.107	38.959	888.31	359.33
36.119	38.992	907.43	362.69
36.131	39.025	926.55	366.11
36.143	39.058	945.67	369.59
36.155	39.091	964.79	373.12
36.167	39.125	983.9	376.72
36.179	39.158	1003.	380.37
36.19	39.192	1022.1	384.07
36.202	39.225	1041.3	387.83
36.214	39.259	1060.4	391.64
36.226	39.293	1079.5	395.5
36.238	39.327	1098.6	399.41
36.25	39.361	1117.7	403.37
36.262	39.395	1136.8	407.38
36.274	39.429	1156.	411.43
36.286	39.463	1175.1	415.54
36.298	39.497	1194.2	419.68
36.31	39.532	1213.3	423.88
36.322	39.566	1232.4	428.11
36.334	39.601	1251.6	432.4
36.346	39.636	1270.7	436.72
36.358	39.671	1289.8	441.09
36.37	39.706	1308.9	445.5
36.382	39.741	1328.	449.95
36.394	39.776	1347.2	454.44
36.406	39.811	1366.3	458.98
36.418	39.846	1385.4	463.55
36.43	39.882	1404.5	468.17
36.442	39.917	1423.6	472.82
36.454	39.953	1442.8	477.52
36.466	39.989	1461.9	482.25
36.478	40.025	1481.	487.02
36.49	40.06	1500.1	491.84
36.502	40.097	1519.2	496.69

36.514	40.133	1538.3	501.57
36.526	40.169	1557.5	506.5
36.538	40.205	1576.6	511.46
36.55	40.242	1595.7	516.46
36.562	40.278	1614.8	521.49
36.574	40.315	1633.9	526.57
36.586	40.352	1653.1	531.68
36.598	40.389	1672.2	536.82
36.61	40.425	1691.3	542.
36.622	40.463	1710.4	547.22
36.634	40.5	1729.5	552.47
36.646	40.537	1748.7	557.76
36.658	40.574	1767.8	563.08
36.67	40.612	1786.9	568.44
36.682	40.649	1806.	573.83
36.694	40.687	1825.1	579.26
36.706	40.725	1844.3	584.71
36.718	40.763	1863.4	590.21
36.73	40.801	1882.5	595.74
36.741	40.839	1901.6	601.3
36.753	40.877	1920.7	606.89
36.765	40.915	1939.8	612.52
36.777	40.954	1959.	618.18
36.789	40.992	1978.1	623.87
36.801	41.031	1997.2	629.6
36.813	41.07	2016.3	635.36
36.825	41.108	2035.4	641.15
36.837	41.147	2054.6	646.97
36.849	41.187	2073.7	652.83
36.861	41.226	2092.8	658.71
36.873	41.265	2111.9	664.63
36.885	41.304	2131.	670.58
36.897	41.344	2150.2	676.56
36.909	41.383	2169.3	682.57
36.921	41.423	2188.4	688.62
36.933	41.463	2207.5	694.69
36.945	41.503	2226.6	700.79
36.957	41.543	2245.8	706.93
36.969	41.583	2264.9	713.09
36.981	41.623	2284.	719.29
36.993	41.664	2303.1	725.51
37.005	41.704	2322.2	731.77
37.017	41.745	2341.3	738.05
37.029	41.785	2360.5	744.37
37.041	41.826	2379.6	750.71
37.052	41.864	2397.4	756.65
37.062	41.899	2398.7	760.62
37.072	41.933	2400.	764.49
37.086	41.979	2401.7	769.59
37.101	42.031	2403.7	775.18
37.117	42.086	2405.7	781.03
37.133	42.144	2407.9	787.05
37.15	42.204	2410.1	793.23
37.168	42.267	2412.4	799.55
37.187	42.332	2414.8	806.07
37.206	42.401	2417.3	812.84
37.227	42.474	2420.	819.85
37.248	42.55	2422.8	827.15
37.271	42.632	2425.7	834.77
37.295	42.719	2428.8	842.75
37.32	42.811	2432.1	851.11

37.347	42.908	2435.5	859.74
37.374	43.008	2439.	868.5
37.402	43.111	2442.6	877.36
37.43	43.217	2446.3	886.29
37.459	43.325	2450.	895.28
37.489	43.435	2453.8	904.29
37.518	43.547	2457.6	913.32
37.548	43.661	2461.5	922.34
37.578	43.776	2465.4	931.33
37.608	43.892	2469.3	940.26
37.638	44.01	2473.2	949.13
37.668	44.129	2477.1	957.95
37.699	44.249	2481.	966.72
37.729	44.37	2485.	975.47
37.76	44.493	2488.9	984.18
37.791	44.618	2492.9	992.88
37.822	44.744	2497.	1001.6
37.854	44.873	2501.	1010.2
37.886	45.004	2505.1	1018.9
37.918	45.136	2509.3	1027.6
37.95	45.271	2513.5	1036.3
37.983	45.409	2517.7	1045.
38.016	45.549	2522.	1053.8
38.035	45.629	2524.4	1058.7
38.053	45.71	2526.8	1063.6
38.064	45.753	2526.2	1066.1
38.074	45.796	2525.9	1068.5
38.088	45.856	2525.1	1071.8
38.122	46.005	2523.	1079.8
38.157	46.157	2520.8	1087.8
38.192	46.312	2518.6	1095.7
38.228	46.471	2516.3	1103.6
38.264	46.633	2514.	1111.4
38.301	46.8	2511.7	1119.3
38.338	46.97	2509.3	1127.1
38.376	47.144	2506.9	1134.9
38.415	47.323	2504.5	1142.6
38.454	47.508	2502.	1150.4
38.494	47.697	2499.5	1158.2
38.535	47.892	2496.9	1166.
38.577	48.093	2494.2	1173.8
38.619	48.301	2491.5	1181.6
38.663	48.516	2488.8	1189.4
38.708	48.739	2486.	1197.3
38.754	48.97	2483.1	1205.2
38.801	49.21	2480.1	1213.1
38.849	49.46	2477.	1221.1
38.899	49.72	2473.9	1229.1
38.95	49.992	2470.6	1237.2
39.003	50.277	2467.3	1245.3
39.055	50.56	2464.	1253.
39.065	50.615	2461.1	1254.1
39.075	50.67	2456.5	1255.1
39.09	50.753	2448.5	1256.5
39.125	50.95	2429.5	1259.6
39.16	51.148	2410.3	1262.4
39.195	51.348	2391.2	1264.9
39.231	51.55	2372.1	1267.2
39.266	51.755	2353.	1269.3
39.301	51.962	2333.9	1271.2
39.336	52.171	2314.7	1272.8

39.371	52.383	2295.6	1274.3
39.406	52.597	2276.5	1275.6
39.441	52.813	2257.4	1276.7
39.477	53.033	2238.3	1277.6
39.512	53.254	2219.2	1278.3
39.547	53.479	2200.	1278.9
39.582	53.707	2180.9	1279.3
39.617	53.937	2161.8	1279.5
39.652	54.17	2142.7	
39.687	54.407	2123.7	1279.4
39.722	54.646	2106.6	1279.1
39.758	54.889	2090.9	1278.7
39.793	55.135	2076.	1278.1
39.828	55.385	2061.7	1277.4
39.863	55.638	2047.2	1276.5
39.898	55.894	2033.	1275.5
39.933	56.154	2021.1	1274.3
39.968	56.418	2009.1	1273.
40.004	56.686	1997.9	1271.6
40.039	56.958	1986.9	1270.
40.056	57.095	1981.4	1269.2
40.066	57.173	1978.6	1268.8
40.076	57.252	1975.5	1268.5
40.1	57.44	1969.	1267.7
40.148	57.827	1957.1	1266.1
40.199	58.246	1944.7	1264.2
40.25	58.681	1934.6	1262.1
40.301	59.126	1924.7	1259.8
40.353	59.582	1915.1	1257.4
40.404	60.048	1907.3	1254.9
40.456	60.525	1899.7	1252.1
40.507	61.013	1893.4	1249.3
40.559	61.513	1887.2	1246.3
40.61	62.025	1881.8	1243.1
40.661	62.549	1876.4	1239.8
40.713	63.086	1871.8	1236.3
40.764	63.635	1867.2	1232.7
40.816	64.198	1862.3	1228.9
40.867	64.774	1857.1	1225.
40.919	65.364	1851.4	1221.
40.97	65.968	1845.5	1216.9
41.021	66.587	1839.2	1212.6
41.058	67.031	1834.7	1209.5
41.068	67.154	1833.4	1208.7
41.078	67.277	1832.1	1208.
41.108	67.653	1828.2	1205.7
41.172	68.48	1819.3	1200.8
41.24	69.377	1809.4	1195.6
41.31	70.343	1798.8	1190.
41.382	71.361	1787.4	1184.3
41.454	72.412	1775.7	1178.4
41.526	73.496	1763.7	1172.4
41.598	74.614	1751.4	1166.3
41.67	75.766	1738.9	1160.1
41.742	76.954	1726.2	1153.8
41.814	78.177	1713.2	1147.4
41.886	79.436	1700.1	1140.9
41.958	80.731	1686.8	1134.2
42.03	82.063	1673.5	1127.5
42.059	82.615	1668.	1124.8
42.069	82.803	1666.1	1123.9

42.079	82.993	1664.2	1123.
42.109	83.566	1658.6	1120.4
42.14	84.149	1652.8	1117.8
42.225	85.847	1636.6	1110.4
42.312	87.645	1619.8	1102.8
42.403	89.556	1602.3	1094.9
42.495	91.569	1584.5	1086.8
42.588	93.677	1566.4	1078.6
42.683	95.876	1548.1	1070.2
42.779	98.161	1529.7	1061.7
42.875	100.53	1511.2	1053.1
42.972	102.97	1492.6	1044.4
43.061	105.25	1475.8	1036.4
43.071	105.51	1473.9	1035.5
43.081	105.77	1472.	1034.7
43.111	106.56	1466.3	1032.1
43.16	107.86	1457.	1027.9
43.26	110.58	1438.2	1019.5
43.36	113.36	1419.5	1011.1
43.46	116.2	1401.	1002.8
43.56	119.11	1382.7	994.5
43.661	122.09	1364.6	986.21
43.761	125.12	1346.7	977.94
43.861	128.21	1329.	969.69
43.961	131.36	1311.6	961.44
44.062	134.59	1294.2	953.14
44.072	134.91	1292.5	952.35
44.082	135.23	1290.8	951.56
44.112	136.2	1285.7	949.2
44.173	138.2	1275.2	944.43
44.274	141.52	1258.5	936.72
44.374	144.88	1242.	929.09
44.474	148.29	1225.7	921.53
44.574	151.74	1209.7	914.01
44.674	155.24	1193.9	906.55
44.774	158.77	1178.4	899.13
44.874	162.35	1163.3	891.75
44.975	165.95	1148.3	884.41
45.063	169.18	1135.3	877.93
45.073	169.54	1133.8	877.22
45.083	169.9	1132.3	876.51
45.113	171.	1127.9	874.41
45.193	173.93	1118.1	868.88
45.293	177.63	1113.6	862.02
45.394	181.35	1109.1	855.24
45.494	185.1	1104.8	848.53
45.594	188.86	1100.9	841.88
45.694	192.64	1097.1	835.29
45.794	196.43	1093.7	828.75
45.894	200.23	1090.2	822.26
45.994	204.04	1086.8	815.81
46.065	206.71	1084.2	811.3
46.075	207.09	1083.9	810.68
46.085	207.47	1083.5	810.05
46.115	208.62	1082.4	808.2
46.205	212.05	1079.	802.7
46.305	215.86	1075.1	796.67
46.405	219.68	1071.	790.71
46.505	223.49	1066.9	784.82
46.606	227.31	1062.6	778.98
46.706	231.12	1058.1	773.21

46.806	234.93	1053.6	767.48
46.906	238.74	1049.	761.8
47.006	242.54	1044.3	756.17
47.066	244.82	1041.4	752.8
47.076	245.2	1041.	752.25
47.086	245.57	1040.5	751.71
47.116	246.71	1039.1	750.08
47.206	250.12	1034.7	745.26
47.307	253.89	1029.7	739.98
47.407	257.65	1024.7	734.76
47.507	261.4	1019.6	729.6
47.607	265.14	1014.4	724.49
47.707	268.86	1009.2	719.44
47.807	272.56	1004.	714.43
47.907	276.25	998.68	709.47
48.008	279.91	993.33	704.55
48.068	282.11	990.11	701.62
48.078	282.47	989.57	701.14
48.088	282.84	989.04	700.66
48.118	283.93	987.42	699.24
48.208	287.2	982.54	695.02
48.308	290.8	977.09	690.4
48.408	294.38	971.61	685.83
48.508	297.94	966.1	681.32
48.608	301.48	960.58	676.86
48.708	304.99	955.04	672.44
48.809	308.47	949.48	668.07
48.909	311.93	943.91	663.74
49.009	315.36	938.32	659.45
49.069	317.41	934.97	656.89
49.079	317.75	934.42	656.47
49.089	318.09	933.86	656.05
49.119	319.11	932.18	654.81
49.209	322.14	927.15	651.12
49.309	325.49	921.55	647.08
49.409	328.8	915.96	643.09
49.51	332.09	910.37	639.15
49.61	335.34	904.78	635.25
49.71	338.56	899.2	631.39
49.81	341.75	893.63	627.57
49.91	344.91	888.07	623.79
50.01	348.03	882.52	620.04
50.07	349.9	879.19	617.81
50.08	350.21	878.64	617.44
50.09	350.52	878.08	617.08
50.12	351.44	876.42	615.99
50.211	354.19	871.46	612.77
50.311	357.22	865.95	609.24
50.411	360.21	860.47	605.75
50.511	363.17	855.	602.31
50.611	366.09	849.56	598.9
50.711	368.98	844.14	595.53
50.811	371.83	838.74	592.2
50.912	374.65	833.37	588.9
51.012	377.43	828.02	585.63
51.072	379.09	824.83	583.68
51.082	379.36	824.29	583.36
51.092	379.64	823.76	583.04
51.122	380.46	822.17	582.09
51.212	382.9	817.41	579.28
51.312	385.58	812.14	576.2

51.412	388.22	806.91	573.15
51.512	390.83	801.71	570.15
51.613	393.4	796.54	567.17
51.713	395.94	791.4	564.23
51.813	398.44	786.29	561.33
51.913	400.9	781.22	558.45
52.013	403.33	776.18	555.6
52.073	404.77	773.16	553.9
52.083	405.01	772.66	553.62
52.093	405.25	772.16	553.34
52.123	405.97	770.66	552.52
52.213	408.09	766.19	550.07
52.314	410.41	761.25	547.38
52.414	412.7	756.35	544.73
52.514	414.95	751.49	542.11
52.614	417.17	746.66	539.52
52.714	419.35	741.87	536.96
52.814	421.23	737.12	534.43
52.914	418.98	732.4	531.93
53.015	416.73	727.72	529.45
53.075	415.37	724.93	527.98
53.085	414.35	724.46	527.65
53.095	413.33	724.	527.33
53.125	410.26	722.61	526.32
53.215	401.05	718.47	523.11
53.315	390.81	713.91	519.39
53.415	380.58	709.38	515.51
53.515	370.35	704.9	511.5
53.615	360.11	700.45	507.37
53.716	349.88	696.04	503.12
53.816	339.65	691.66	498.75
53.916	329.41	687.31	494.29
54.016	319.18	683.	489.72
54.076	313.04	680.42	486.95
54.086	312.01	679.99	486.48
54.096	310.99	679.56	486.02
54.126	307.91	678.27	484.62
54.216	298.69	674.43	480.34
54.316	288.44	670.18	475.49
54.417	278.19	665.95	470.57
54.517	267.95	661.72	465.56
54.617	257.7	657.49	460.47
54.717	247.45	653.27	455.31
54.817	237.2	649.04	450.08
54.917	226.95	644.8	444.77
55.017	216.7	640.55	439.4
55.078	210.56	637.99	436.16
55.088	210.35	637.57	435.69
55.098	210.15	637.14	435.24
55.128	209.53	635.86	433.89
55.218	207.7	632.	429.99
55.318	205.66	627.69	425.79
55.418	203.61	623.36	421.71
55.518	201.57	618.99	417.74
55.618	199.53	614.6	413.86
55.718	197.49	610.17	410.07
55.819	195.45	605.72	406.36
55.919	193.41	601.23	402.73
56.019	191.36	596.71	399.18
56.079	190.14	593.99	397.07
56.089		593.54	396.74

56.099	190.13	593.09	396.41
56.129	190.12	591.72	395.43
56.219	190.1	587.61	392.61
56.319	190.07	583.02	389.59
56.419	190.04	578.41	386.67
56.52	190.01	573.78	383.85
56.62	189.98	569.36	381.12
56.72	189.95	566.17	378.47
56.82	189.92	563.19	375.91
56.92	189.9	560.25	373.43
57.02	189.87	557.39	371.02
57.08		555.76	369.6
57.09		555.49	369.36
57.1		555.22	369.13
57.13		554.42	368.43
57.22		551.99	366.39
57.321		549.45	364.19
57.421		547.05	362.06
57.521		544.64	359.98
57.621		542.39	357.97
57.721		540.24	356.01
57.821		538.16	354.1
57.921		536.18	352.24
58.022		534.29	350.43
58.082	189.85	533.2	349.36
58.092		533.02	349.19
58.102		532.85	349.01
58.132		532.33	348.48
58.222		530.79	346.94
58.322		529.18	345.26
58.422		527.66	343.63
58.522		526.21	342.04
58.622		524.85	340.49
58.723		523.6	338.97
58.823		522.45	337.49
58.923		521.37	336.04
59.023		520.37	334.62
59.083		519.82	333.79
59.093	189.8	519.73	333.64
59.103	189.75	519.64	333.5
59.133	189.6	519.39	333.07
59.223	189.14	518.65	331.79
59.323	188.64	517.9	330.38
59.424	188.14	517.22	328.99
59.524	187.63	516.59	327.62
59.624	187.13	516.04	326.26
59.724	186.62	515.55	324.93
59.824	186.12	515.07	323.61
59.924	185.61	514.64	322.3
60.024	185.11	514.22	321.01
60.084	184.81	513.97	320.24
60.095	184.75	513.93	320.11
60.105	184.7	513.89	319.99
60.135	184.54	513.77	319.6
60.225	184.06	513.39	318.47
60.325	183.53	512.97	317.22
60.425	182.99	512.53	315.98
60.525	182.46	512.1	314.76
60.625	181.92	511.66	313.54
60.725	181.39	511.22	312.35
60.826	180.85	510.76	311.16

60.926	180.32	510.3	309.99
61.026	179.79	509.82	308.82
61.086	179.47	509.53	308.13
61.096	179.42	509.49	308.01
61.106	179.37	509.44	307.9
61.136	179.24	509.29	307.56
61.226	178.82	508.84	306.54
61.326	178.36	508.34	305.42
61.426	177.9	507.82	304.31
61.527	177.44	507.3	303.22
61.627	176.98	506.76	302.13
61.727	176.52	506.21	301.06
61.827	176.06	505.65	300.
61.927	175.6	505.09	298.95
62.027	175.14	504.52	297.91
62.087	174.87	504.17	297.28
62.097	174.83	504.11	297.18
62.107	174.79	504.05	297.08
62.137	174.67	503.88	296.77
62.227	174.33	503.34	295.86
62.328	173.95	502.73	294.86
62.428	173.57	502.11	293.87
62.528	173.18	501.49	292.89
62.628	172.8	500.86	291.92
62.728	172.42	500.23	290.96
62.828	172.04	499.59	290.01
62.928	171.66	498.93	289.07
63.029	171.27	498.27	288.14
63.089	171.04	497.86	287.58
63.099	171.01	497.8	287.49
63.109	170.98	497.73	287.4
63.139	170.89	497.53	287.12
63.229	170.62	496.93	286.31
63.329	170.31	496.24	285.42
63.429	170.	495.55	284.53
63.529	169.7	494.86	283.65
63.629	169.39	494.16	282.79
63.73	169.08	493.45	281.93
63.83	168.78	492.74	281.08
63.93	168.47	492.28	280.24
64.03	168.16	491.81	279.41
64.09	167.98	491.53	278.91
64.1	167.96	491.48	278.83
64.11	167.93	491.43	278.74
64.14	167.86	491.29	278.5
64.23	167.65	490.85	277.77
64.33	167.41	490.35	276.98
64.431	167.17	489.84	276.19
64.531	166.93	489.32	275.4
64.631	166.69	488.8	274.63
64.731	166.46	488.26	273.86
64.831	166.22	487.71	273.11
64.931	165.98	487.15	272.36
65.031	165.74	486.58	271.61
65.092	165.6	486.24	271.17
65.102	165.58	486.18	271.09
65.112	165.56	486.13	271.02
65.142	165.51	485.95	270.8
65.232	165.35	485.43	270.15
65.332	165.17	484.84	269.44
65.432	164.99	484.24	268.74

65.532	164.81	483.63	268.04
65.632	164.63	483.01	267.35
65.732	164.45	482.39	266.67
65.833	164.27	481.75	265.99
65.933	164.09	481.11	265.32
66.033	163.92	480.47	264.66
66.093	163.81	480.08	264.26
66.103	163.79	480.01	264.19
66.113	163.78	479.95	264.13
66.143	163.74	479.75	263.93
66.233	163.63	479.16	263.35
66.333	163.49	478.49	262.72
66.433	163.36	477.82	262.09
66.534	163.23	477.14	261.47
66.634	163.1	476.45	260.85
66.734	162.97	475.76	260.24
66.834	162.84	475.07	259.63
66.934	162.71	474.36	259.04
67.034	162.58	473.66	258.44
67.094	162.5	473.23	258.09
67.104	162.49	473.16	258.03
67.114		473.09	257.97
67.144	162.46	472.87	257.8
67.235	162.37	472.23	257.28
67.335	162.28	471.5	256.71
67.435	162.19	470.77	256.15
67.535	162.1	470.04	255.59
67.635	162.01	469.3	255.04
67.735	161.91	468.56	254.49
67.835	161.82	467.82	253.95
67.936	161.73	467.07	253.41
68.036	161.64	466.31	252.88
68.096	161.58	465.86	252.56
68.106		465.78	252.51
68.116	161.57	465.71	252.46
68.146	161.55	465.48	252.3
68.236	161.49	464.79	251.83
68.336	161.43	464.03	251.32
68.436	161.37	463.26	250.82
68.536	161.3	462.48	250.31
68.637	161.24	461.71	249.82
68.737	161.18	460.93	249.33
68.837	161.11	460.15	248.84
68.937	161.05	459.36	248.36
69.037	160.99	458.57	247.88
69.097	160.95	458.1	247.59
69.107		458.02	247.54
69.117	160.94	457.94	247.5
69.147	160.93	457.7	247.36
69.237	160.89	456.99	246.94
69.338	160.85	456.19	246.47
69.438	160.81	455.39	246.02
69.538	160.77	454.59	245.56
69.638	160.72	453.79	245.11
69.738	160.68	452.98	244.67
69.838	160.64	452.17	244.23
69.938	160.6	451.36	243.79
70.039	160.56	450.55	243.36
70.099	160.53	450.06	243.1
70.109		449.98	243.06
70.119		449.9	243.01

70.149	160.52	449.66	242.88
70.239	160.49	448.92	242.5
70.339	160.47	448.11	242.08
70.439	160.44	447.29	241.67
70.539	160.41	446.47	241.26
70.639	160.38	445.65	240.85
70.739	160.36	444.83	240.44
70.84	160.33	444.	240.04
70.94	160.3	443.18	239.64
71.04	160.28	442.35	239.25
71.1	160.26	441.85	239.01

FIGURE 6
Model (A4) > Transient Thermal (A5) > Solution (A6) > Total Heat Flux

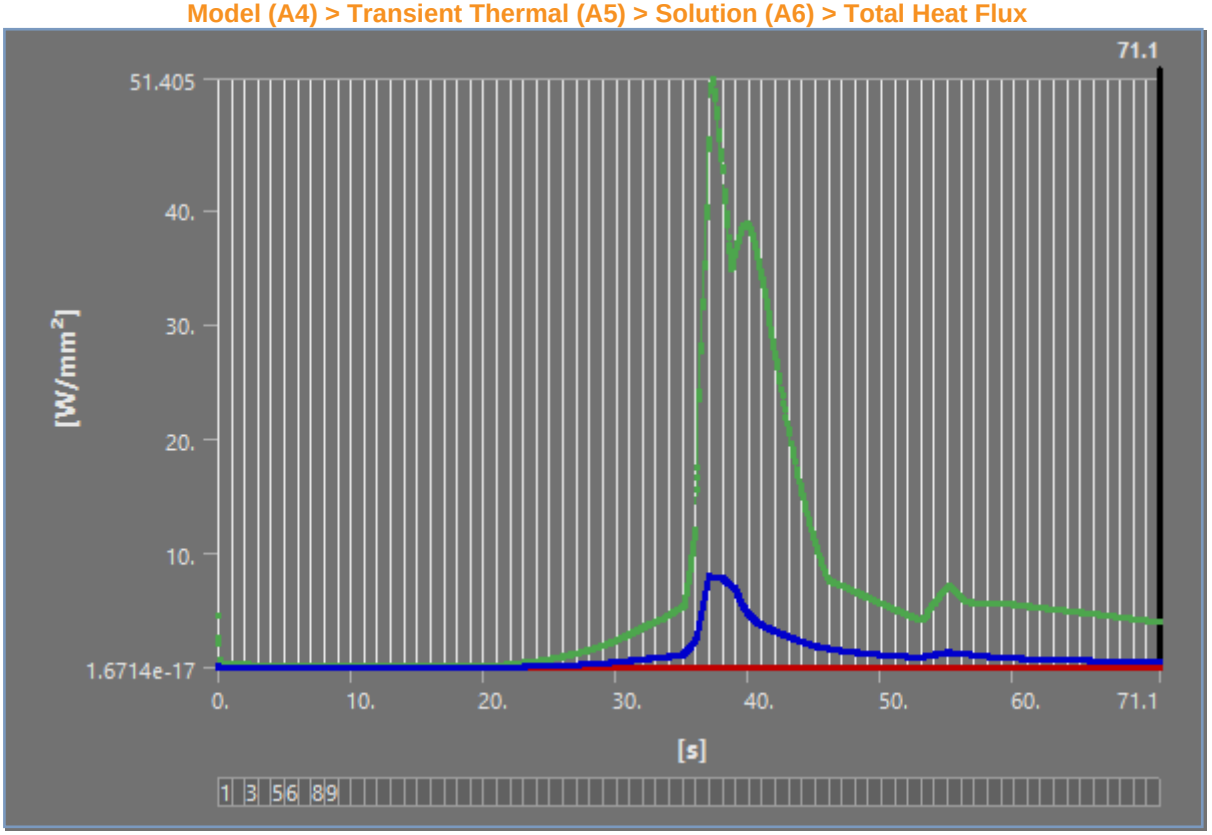


TABLE 24
Model (A4) > Transient Thermal (A5) > Solution (A6) > Total Heat Flux

Time [s]	Minimum [W/mm²]	Maximum [W/mm²]	Average [W/mm²]
1.0014e-002	4.0441e-017	4.4697	0.14228
1.7163e-002	8.4816e-017	2.6255	0.11058
2.1855e-002	3.2296e-015	2.0249	9.7321e-002
2.6548e-002	1.289e-014	1.659	8.8612e-002
3.7797e-002	5.3015e-015	1.2395	7.8431e-002
5.6425e-002	1.1626e-015	0.95227	7.0026e-002
8.5663e-002	1.6259e-016	0.76583	6.4347e-002
0.13041	1.0704e-016	0.62116	6.049e-002
0.19908	1.0125e-016	0.50151	5.8161e-002
0.29922	1.2253e-015	0.4051	5.6884e-002
0.39936	2.2274e-015	0.38321	5.6328e-002
0.4995	2.7466e-016	0.40705	5.602e-002
0.59964	4.6169e-017	0.41641	5.5785e-002
0.69978	3.0811e-017	0.41711	5.5553e-002
0.79992	3.4072e-017	0.41263	5.5301e-002
0.90006	3.069e-017	0.40509	5.5022e-002
1.0014	4.2673e-017	0.39569	5.4712e-002
1.0114	2.6031e-017	0.39485	5.4696e-002
1.0214	4.7929e-017	0.3939	5.4672e-002

1.0515	3.3607e-017	0.39088	5.4584e-002
1.1416	3.86e-017	0.38124	5.428e-002
1.2417	4.0953e-017	0.3703	5.3918e-002
1.3419		0.35939	5.354e-002
1.442		0.34866	5.3149e-002
1.5422		0.33821	5.2749e-002
1.6423		0.32809	5.2341e-002
1.7424		0.31835	5.1929e-002
1.8426		0.30899	5.1513e-002
1.9427		0.30001	5.1096e-002
2.0028	4.8555e-017	0.29476	5.0845e-002
2.0128	2.6682e-017	0.29387	5.0804e-002
2.0228	3.4981e-017	0.293	5.0763e-002
2.0529	2.4871e-017	0.29041	5.0638e-002
2.143	2.689e-017	0.28295	5.0262e-002
2.2431	3.1765e-017	0.27504	4.9844e-002
2.3433	4.7142e-017	0.26748	4.9428e-002
2.4434	4.6921e-017	0.26026	4.9015e-002
2.5436		0.25336	4.8605e-002
2.6437		0.24677	4.8199e-002
2.7438		0.24046	4.7797e-002
2.844		0.23443	4.74e-002
2.9441		0.22866	4.7006e-002
3.0042	4.7438e-017	0.22528	4.6772e-002
3.0142	3.9462e-017		4.6734e-002
3.0242	2.8397e-017	0.2257	4.6695e-002
3.0543	3.6658e-017	0.22689	4.6578e-002
3.1444	2.689e-017	0.23019	4.6232e-002
3.2445	3.1765e-017	0.23354	4.5853e-002
3.3447	4.7142e-017	0.23662	4.5478e-002
3.4448	4.6921e-017	0.23943	4.5108e-002
3.545		0.242	4.4744e-002
3.6451		0.24434	4.4385e-002
3.7452		0.24647	4.4031e-002
3.8454		0.2484	4.3682e-002
3.9455		0.25014	4.3339e-002
4.0056	4.7438e-017	0.25113	4.3134e-002
4.0156	2.4915e-017	0.2513	4.31e-002
4.0256	2.5097e-017	0.25147	4.3067e-002
4.0557	2.6159e-017	0.25194	4.2965e-002
4.1458	4.4212e-017	0.25321	4.2664e-002
4.2459	4.318e-017	0.25447	4.2335e-002
4.3461		0.25558	4.2011e-002
4.4462		0.25655	4.1692e-002
4.5464		0.2574	4.1378e-002
4.6465		0.25813	4.1068e-002
4.7466		0.25875	4.0763e-002
4.8468		0.25926	4.0463e-002
4.9469		0.25968	4.0168e-002
5.007	1.9413e-017	0.2599	3.9992e-002
5.017	2.8127e-017	0.25994	3.9963e-002
5.027	2.7996e-017	0.25998	3.9934e-002
5.0571	2.5252e-017	0.26008	3.9847e-002
5.1472	3.4441e-017	0.26029	3.9588e-002
5.2474	3.6038e-017	0.26046	3.9306e-002
5.3475	2.8288e-017	0.26054	3.9027e-002
5.4477		0.26056	3.8753e-002
5.5478		0.26052	3.8483e-002
5.648		0.26042	3.8217e-002
5.7481		0.26026	3.7955e-002
5.8483		0.26005	3.7697e-002

5.9484		0.25979	3.7442e-002
6.0085	2.6439e-017	0.25961	3.7291e-002
6.0185	2.9585e-017	0.25958	3.7266e-002
6.0285	3.6027e-017	0.25955	3.7241e-002
6.0586	3.2104e-017	0.25946	3.7166e-002
6.1487	2.689e-017	0.25914	3.6943e-002
6.2488	3.1765e-017	0.25875	3.67e-002
6.349	4.7142e-017	0.25832	3.646e-002
6.4491	4.6921e-017	0.25786	3.6223e-002
6.5493		0.25737	3.599e-002
6.6494		0.25685	3.576e-002
6.7495		0.2563	3.5534e-002
6.8497		0.25573	3.531e-002
6.9498		0.25514	3.509e-002
7.0099	4.0847e-017	0.25477	3.4959e-002
7.0199	2.7158e-017	0.25471	3.4937e-002
7.0299	2.7167e-017	0.25464	3.4915e-002
7.06	2.0347e-017	0.25445	3.485e-002
7.1501	3.5618e-017	0.25388	3.4657e-002
7.2502	4.1292e-017	0.25322	3.4446e-002
7.3504		0.25254	3.4237e-002
7.4505		0.25186	3.4031e-002
7.5507		0.25115	3.3828e-002
7.6508		0.25044	3.3628e-002
7.7509		0.24972	3.343e-002
7.8511		0.24898	3.3235e-002
7.9512		0.24824	3.3043e-002
8.0113	2.9296e-017	0.24779	3.2928e-002
8.0213	2.9977e-017	0.24771	3.2909e-002
8.0313	4.4146e-017	0.24763	3.289e-002
8.0614	4.1528e-017	0.2474	3.2833e-002
8.1515	2.9392e-017	0.24671	3.2664e-002
8.2516	3.5646e-017	0.24594	3.2478e-002
8.3518	3.6558e-017	0.24516	3.2295e-002
8.4519		0.24438	3.2115e-002
8.5521		0.2436	3.1936e-002
8.6522		0.24281	3.176e-002
8.7523		0.24202	3.1586e-002
8.8525		0.24122	3.1414e-002
8.9526		0.24043	3.1244e-002
9.0127	4.3122e-017	0.23995	3.1143e-002
9.0227	2.1073e-017	0.23986	3.1126e-002
9.0327	1.6714e-017	0.23978	3.1109e-002
9.0628	2.531e-017	0.23954	3.1059e-002
9.1529	3.5438e-017	0.23881	3.0909e-002
9.253	4.1292e-017	0.23801	3.0745e-002
9.3532		0.23721	3.0583e-002
9.4533		0.2364	3.0423e-002
9.5535		0.2356	3.0265e-002
9.6536		0.2348	3.0108e-002
9.7537		0.234	2.9954e-002
9.8539		0.2332	2.98e-002
9.954		0.2324	2.9649e-002
10.014	2.9296e-017	0.23192	2.9559e-002
10.024	2.9977e-017	0.23183	2.9544e-002
10.034	4.4146e-017	0.23175	2.9529e-002
10.064	4.1528e-017	0.23151	2.9484e-002
10.154	2.9392e-017	0.23079	2.9351e-002
10.254	3.5646e-017	0.22999	2.9204e-002
10.355	3.6558e-017	0.2292	2.9059e-002
10.455		0.22841	2.8916e-002

10.555		0.22762	2.8774e-002
10.655		0.22684	2.8634e-002
10.755		0.22605	2.8495e-002
10.855		0.22527	2.8358e-002
10.955		0.2245	2.8222e-002
11.016	4.3122e-017	0.22403	2.8141e-002
11.026	2.1073e-017	0.22395	2.8127e-002
11.036	1.6714e-017	0.22387	2.8114e-002
11.066	2.531e-017	0.22364	2.8073e-002
11.156	3.5438e-017	0.22294	2.7953e-002
11.256	4.1292e-017	0.22217	2.7822e-002
11.356		0.2214	2.7691e-002
11.456		0.22064	2.7562e-002
11.556		0.21988	2.7434e-002
11.656		0.21913	2.7307e-002
11.757		0.21838	2.7181e-002
11.857		0.21763	2.7057e-002
11.957		0.21688	2.6934e-002
12.017	2.9296e-017	0.21644	2.686e-002
12.027	2.765e-017	0.21636	2.6848e-002
12.037	2.0395e-017	0.21628	2.6836e-002
12.067	2.3479e-017	0.21606	2.6799e-002
12.157	3.5618e-017	0.21539	2.669e-002
12.257	4.1292e-017	0.21465	2.6571e-002
12.357		0.21392	2.6452e-002
12.458		0.21319	2.6334e-002
12.558		0.21247	2.6218e-002
12.658		0.21175	2.6102e-002
12.758		0.21103	2.5988e-002
12.858		0.21032	2.5875e-002
12.958		0.20961	2.5762e-002
13.018	2.9296e-017	0.20918	2.5695e-002
13.028	2.9977e-017	0.20911	2.5684e-002
13.038	4.4146e-017	0.20903	2.5673e-002
13.068	4.1528e-017	0.20882	2.5639e-002
13.158	2.9392e-017	0.20818	2.554e-002
13.259	3.5646e-017	0.20748	2.543e-002
13.359	3.6558e-017	0.20679	2.5322e-002
13.459		0.20609	2.5214e-002
13.559		0.2054	2.5107e-002
13.659		0.20472	2.5001e-002
13.759		0.20403	2.4896e-002
13.859		0.20335	2.4792e-002
13.96		0.20268	2.4689e-002
14.02	4.3122e-017	0.20227	2.4627e-002
14.03	2.1073e-017	0.2022	2.4617e-002
14.04	1.6714e-017	0.20213	2.4607e-002
14.07	2.531e-017	0.20193	2.4576e-002
14.16	3.5438e-017	0.20132	2.4484e-002
14.26	4.1292e-017	0.20065	2.4383e-002
14.36		0.19999	2.4283e-002
14.46		0.19933	2.4184e-002
14.56		0.19867	2.4086e-002
14.661		0.19802	2.3988e-002
14.761		0.19737	2.3891e-002
14.861		0.19672	2.3795e-002
14.961		0.19608	2.3699e-002
15.021	2.9296e-017	0.19569	2.3642e-002
15.031	2.765e-017	0.19562	2.3633e-002
15.041	2.0395e-017	0.19556	2.3623e-002
15.071	2.3479e-017	0.19536	2.3595e-002

15.161	3.5618e-017	0.19479	2.351e-002
15.261	4.1292e-017	0.19415	2.3417e-002
15.362		0.19352	2.3324e-002
15.462		0.19289	2.3232e-002
15.562		0.19226	2.314e-002
15.662		0.19163	2.305e-002
15.762		0.19101	2.296e-002
15.862		0.19039	2.287e-002
15.962		0.18978	2.2781e-002
16.022	2.9296e-017	0.18941	2.2728e-002
16.033	2.5124e-017	0.18935	2.2719e-002
16.043	2.7669e-017	0.18928	2.271e-002
16.073	3.7258e-017	0.1891	2.2684e-002
16.163	2.8057e-017	0.18855	2.2605e-002
16.263	4.958e-017	0.18794	2.2518e-002
16.363	4.939e-017	0.18733	2.2432e-002
16.463	4.2449e-017	0.18673	2.2346e-002
16.563		0.18613	2.2261e-002
16.663		0.18553	2.2176e-002
16.764		0.18494	2.2092e-002
16.864		0.18435	2.2008e-002
16.964		0.18376	2.1925e-002
17.024	2.2116e-017	0.18341	2.1876e-002
17.034	7.6707e-013	0.18334	2.1871e-002
17.044	5.6435e-013	0.18328	2.1865e-002
17.074	1.873e-013	0.18311	2.1847e-002
17.164	9.9149e-015	0.18258	2.1788e-002
17.264	8.775e-015	0.182	2.1723e-002
17.364	2.426e-015	0.18142	2.1658e-002
17.465	2.091e-015	0.18084	2.1593e-002
17.565	2.0605e-015	0.18027	2.1528e-002
17.665	2.0715e-015	0.1797	2.1464e-002
17.765	2.086e-015	0.17914	2.1401e-002
17.865	2.0764e-015	0.17859	2.1338e-002
17.965	2.0888e-015	0.17804	2.1275e-002
18.025	2.0742e-015	0.17771	2.1237e-002
18.035	1.8339e-011	0.17765	2.1307e-002
18.045	1.4044e-011	0.1776	2.136e-002
18.075	4.2401e-012	0.17743	2.1479e-002
18.166	2.4227e-013	0.17695	2.1771e-002
18.266	1.5422e-013	0.17642	2.2074e-002
18.366	1.2681e-013	0.17592	2.2368e-002
18.466	1.2379e-013	0.17545	2.2658e-002
18.566	1.2363e-013	0.17502	2.2947e-002
18.666		0.17465	2.3234e-002
18.766		0.17435	2.352e-002
18.867		0.17412	2.3805e-002
18.967		0.17397	2.4088e-002
19.027		0.17391	2.4258e-002
19.037	1.3086e-011	0.17389	2.4393e-002
19.047	1.9537e-011	0.17388	2.4506e-002
19.077	2.8783e-012	0.17387	2.4786e-002
19.167	1.0191e-012	0.17388	2.5534e-002
19.267	3.7722e-013	0.17398	2.6331e-002
19.367	2.9478e-013	0.17419	2.7115e-002
19.467	2.9046e-013	0.17451	2.7892e-002
19.568	2.9025e-013	0.17496	2.8664e-002
19.668		0.17555	2.9432e-002
19.768		0.1763	3.0196e-002
19.868	2.9026e-013	0.17721	3.0955e-002
19.968	2.9027e-013	0.17828	3.171e-002

20.028		0.17898	3.2162e-002
20.038	3.6771e-011	0.1791	3.2349e-002
20.048	2.3139e-011	0.17922	3.2512e-002
20.078	1.4018e-012	0.17959	3.2943e-002
20.168	5.956e-013	0.18086	3.4133e-002
20.269	5.5465e-013	0.18243	3.5419e-002
20.369	4.6925e-013	0.18417	3.6688e-002
20.469	4.6476e-013	0.1861	3.7947e-002
20.569	4.6454e-013	0.18823	3.9198e-002
20.669	4.6453e-013	0.19057	4.0442e-002
20.769	4.6454e-013	0.19311	4.1678e-002
20.869	4.6455e-013	0.19588	4.2907e-002
20.97	4.6456e-013	0.19886	4.4127e-002
21.03	4.6457e-013	0.20073	4.4857e-002
21.04	4.6251e-011	0.20105	4.5098e-002
21.05	7.3429e-012	0.20137	4.5314e-002
21.08	7.9323e-012	0.20235	4.5898e-002
21.17	9.7314e-013	0.20965	4.7541e-002
21.27	7.4749e-013	0.21952	4.9325e-002
21.37	6.5642e-013	0.22995	5.1089e-002
21.47	6.5159e-013	0.24059	5.284e-002
21.57	6.5136e-013	0.25124	5.458e-002
21.67		0.26178	5.6309e-002
21.771	6.5138e-013	0.27214	5.8028e-002
21.871	6.5139e-013	0.28229	5.9737e-002
21.971	6.514e-013	0.29219	6.1434e-002
22.031	6.5141e-013	0.29805	6.2449e-002
22.041	6.0935e-011	0.29895	6.2749e-002
22.051	2.4621e-011	0.29992	6.3021e-002
22.081	9.1707e-012	0.3029	6.377e-002
22.171	9.2802e-013	0.31296	6.5895e-002
22.271	9.6154e-013	0.32524	6.8209e-002
22.371	8.6191e-013	0.33809	7.05e-002
22.472	8.5662e-013	0.35115	7.2775e-002
22.572	8.5637e-013	0.3642	7.5036e-002
22.672		0.37711	7.7284e-002
22.772	8.5638e-013	0.3898	7.952e-002
22.872	8.5641e-013	0.40224	8.1742e-002
22.972	8.5643e-013	0.4144	8.3951e-002
23.032	8.5644e-013	0.4216	8.5272e-002
23.042	6.3498e-011	0.42272	8.5637e-002
23.052	5.3288e-011	0.42391	8.5973e-002
23.082	1.0729e-011	0.42758	8.6905e-002
23.173	6.6197e-013	0.43987	8.9567e-002
23.273	1.2042e-012	0.45473	9.2471e-002
23.373	1.0926e-012	0.47022	9.5348e-002
23.473	1.0867e-012	0.48594	9.8207e-002
23.573	1.0864e-012	0.50163	0.10105
23.673		0.51715	0.10388
23.773		0.53242	0.10669
23.874		0.5474	0.10949
23.974	1.0865e-012	0.56207	0.11227
24.034		0.57076	0.11393
24.044	5.2504e-011	0.57211	0.11438
24.054	6.1417e-011	0.57355	0.11479
24.084	1.267e-011	0.57801	0.11593
24.174	4.7797e-013	0.593	0.11921
24.274	1.4845e-012	0.61116	0.1228
24.374	1.3569e-012	0.63001	0.12636
24.474	1.3501e-012	0.64909	0.12989
24.575	1.3498e-012	0.66812	0.13341

24.675		0.68696	0.13691
24.775		0.7055	0.14039
24.875	1.3499e-012	0.72371	0.14386
24.975		0.74156	0.14731
25.035		0.75215	0.14937
25.045	4.5477e-011	0.75402	0.14991
25.055	7.1829e-011	0.75583	0.15041
25.085	1.5068e-011	0.76131	0.15181
25.175	4.1529e-013	0.77954	0.15583
25.276	1.8132e-012	0.80138	0.16023
25.376	1.6647e-012	0.82402	0.1646
25.476	1.6568e-012	0.84693	0.16894
25.576	1.6564e-012	0.86978	0.17327
25.676	1.6565e-012	0.8924	0.17758
25.776		0.91469	0.18186
25.876		0.93659	0.18613
25.977	1.6566e-012	0.9581	0.19038
26.037		0.97086	0.19293
26.047	4.3325e-011	0.97312	0.19358
26.057	8.5184e-011	0.97532	0.19418
26.087	1.8048e-011	0.98196	0.19589
26.177	4.774e-013	1.004	0.20081
26.277	2.2045e-012	1.0305	0.20617
26.377	2.0289e-012	1.0578	0.21149
26.477	2.0196e-012	1.0855	0.21678
26.577	2.0191e-012	1.1132	0.22204
26.677		1.1405	0.22727
26.778	2.0192e-012	1.1675	0.23248
26.878		1.194	0.23766
26.978	2.0193e-012	1.2201	0.24281
27.038		1.2352	0.2459
27.048	4.5383e-011	1.2377	0.24672
27.058	1.0197e-010	1.2402	0.24746
27.088	2.1723e-011	1.2477	0.24955
27.178	6.0092e-013	1.2736	0.25552
27.278	2.6742e-012	1.305	0.26204
27.378	2.4644e-012	1.3369	0.26849
27.479	2.4532e-012	1.3692	0.2749
27.579	2.4527e-012	1.4013	0.28129
27.679		1.433	0.28764
27.779		1.4641	0.29397
27.879	2.4528e-012	1.4941	0.30026
27.979		1.5236	0.3065
28.039	2.4529e-012	1.541	0.31024
28.049	4.9451e-011	1.5442	0.31121
28.059	1.2236e-010	1.5473	0.31211
28.089	2.6163e-011	1.5565	0.31464
28.18	7.6208e-013	1.587	0.32189
28.28	3.2378e-012	1.6238	0.32982
28.38	2.9863e-012	1.6621	0.33769
28.48	2.973e-012	1.7008	0.34552
28.58	2.9723e-012	1.7395	0.3533
28.68	2.9724e-012	1.7778	0.36104
28.78		1.8166	0.36862
28.881	2.9725e-012	1.8542	0.37624
28.981	2.9726e-012	1.8907	0.38383
29.041		1.9122	0.38837
29.051	4.9065e-011	1.9159	0.38957
29.061	1.4506e-010	1.9195	0.39067
29.091	3.1228e-011	1.9305	0.39373
29.181	1.0266e-012	1.9679	0.40253

29.281	3.9014e-012	2.0129	0.41216
29.381	3.604e-012	2.0598	0.42173
29.481	3.5882e-012	2.1071	0.43125
29.582	3.5874e-012	2.1544	0.44074
29.682		2.2016	0.45006
29.782	3.5875e-012	2.2478	0.45941
29.882	3.5876e-012	2.2929	0.46873
29.982	3.5877e-012	2.337	0.47799
30.042		2.3631	0.48351
30.052	4.5082e-011	2.3676	0.48491
30.062	1.6529e-010	2.372	0.48621
30.092	3.6252e-011	2.3855	0.48986
30.182	1.6945e-012	2.4306	0.50033
30.283	4.6403e-012	2.4848	0.51176
30.383	4.3033e-012	2.541	0.52308
30.483	4.2854e-012	2.5979	0.53431
30.583	4.2845e-012	2.6537	0.54567
30.683	4.2846e-012	2.7087	0.55678
30.783	4.2847e-012	2.763	0.56779
30.883	4.2848e-012	2.8165	0.57874
30.984	4.2849e-012	2.8659	0.58964
31.044	4.285e-012	2.8956	0.59614
31.054	9.4151e-011	2.8994	0.59769
31.064	1.6734e-010	2.904	0.59915
31.094	3.8964e-011	2.9186	0.6033
31.184	3.8419e-012	2.9692	0.61523
31.284	5.3324e-012	3.0274	0.62824
31.384	4.9974e-012	3.0868	0.64112
31.484	4.9795e-012	3.1419	0.65392
31.584	4.9787e-012	3.1987	0.66661
31.685	4.9789e-012	3.255	0.67922
31.785	4.979e-012	3.3102	0.69174
31.885		3.3679	0.70422
31.985	4.9792e-012	3.4207	0.71654
32.045	4.9793e-012	3.4517	0.7239
32.055	1.815e-010	3.4577	0.72541
32.065	3.7056e-011	3.4635	0.72687
32.095	1.9091e-011	3.4801	0.73109
32.185	6.2395e-012	3.5303	0.74347
32.285	5.641e-012	3.5875	0.75708
32.386	5.43e-012	3.6451	0.77055
32.486	5.4189e-012	3.7041	0.78372
32.586	5.4184e-012	3.7616	0.7969
32.686	5.4185e-012	3.8175	0.80995
32.786	5.4186e-012	3.8726	0.82291
32.886	5.4187e-012	3.9263	0.83576
32.986	5.4189e-012	3.9785	0.84853
33.047		4.0095	0.85624
33.057	4.6697e-011	4.0146	0.85727
33.067	6.0457e-011	4.0197	0.85835
33.097	2.5779e-011	4.0348	0.86168
33.187	2.6228e-012	4.0763	0.87182
33.287	4.8898e-012	4.1184	0.88306
33.387	5.0574e-012	4.1579	0.89424
33.487	5.0666e-012	4.1956	0.90527
33.587	5.0671e-012	4.2321	0.91628
33.687	5.0673e-012	4.3093	0.92722
33.788	5.0675e-012	4.3886	0.93789
33.888		4.4666	0.94861
33.988	5.0677e-012	4.5446	0.95924
34.048		4.5926	0.96552

34.058	1.3077e-010	4.6006	0.96527
34.068	2.2307e-011	4.6085	0.96531
34.098	5.7469e-011	4.6317	0.96607
34.188	6.2578e-012	4.7011	0.96934
34.288	1.5072e-012	4.7784	0.97322
34.388	3.2003e-012	4.8557	0.97715
34.489	3.2472e-012	4.9326	0.98107
34.589	3.2497e-012	5.0073	0.98513
34.689	3.2499e-012	5.0797	0.98899
34.789	3.25e-012	5.1514	0.99282
34.889	3.2501e-012	5.2224	0.99662
34.989		5.2877	1.0005
35.049	3.2502e-012	5.3266	1.0028
35.059	5.995e-009	5.333	1.0292
35.069	4.7641e-009	5.339	1.0499
35.095	1.4854e-009	5.3548	1.0926
35.146	2.1825e-010	5.3864	1.1662
35.197	1.2616e-011	5.4182	1.2356
35.248	4.6474e-011	5.6793	1.3033
35.299	4.0306e-011	6.1233	1.3701
35.35	3.9716e-011	6.5221	1.4363
35.401	3.966e-011	6.8869	1.5023
35.452	3.9655e-011	7.2251	1.5681
35.503		7.5416	1.6337
35.554	3.9656e-011	7.8402	1.6992
35.605		8.1349	1.7645
35.656		8.6111	1.8297
35.707	3.9657e-011	9.0837	1.8948
35.758		9.5512	1.9596
35.809	3.9658e-011	10.013	2.0244
35.86		10.468	2.0889
35.911	3.9659e-011	10.915	2.1532
35.962		11.356	2.2173
36.012	3.966e-011	11.789	2.2811
36.051		12.11	2.3289
36.061	2.5592e-008	12.572	2.4335
36.071	1.5709e-008	14.284	2.5181
36.083	1.8166e-009	15.628	2.6068
36.095	2.1504e-009	16.654	2.6891
36.107	1.1031e-009	17.508	2.7676
36.119	3.7257e-010	18.255	2.8437
36.131	2.4032e-010	18.928	2.918
36.143	1.5483e-010	19.546	2.9912
36.155	2.1511e-010	20.121	3.0633
36.167	1.8262e-010	20.662	3.1347
36.179	1.7286e-010	21.173	3.2055
36.19	1.6992e-010	21.659	3.2759
36.202	1.6904e-010	22.124	3.3458
36.214	1.6877e-010	22.57	3.4155
36.226	1.687e-010	22.999	3.4848
36.238	1.6867e-010	23.413	3.554
36.25		23.812	3.6229
36.262		24.2	3.6917
36.274		24.575	3.7604
36.286		24.94	3.829
36.298		25.294	3.8974
36.31		25.639	3.9658
36.322		25.976	4.0341
36.334		26.304	4.1023
36.346		26.624	4.1705
36.358		26.937	4.2386

36.37		27.269	4.3067
36.382		27.598	4.3747
36.394		27.921	4.4427
36.406		28.236	4.5106
36.418		28.545	4.5785
36.43		28.849	4.6464
36.442		29.146	4.7142
36.454		29.438	4.782
36.466		29.725	4.8498
36.478		30.007	4.9175
36.49		30.285	4.9852
36.502		30.558	5.0529
36.514		30.842	5.1205
36.526		31.133	5.1881
36.538		31.42	5.2556
36.55		31.704	5.3231
36.562		31.984	5.3906
36.574		32.262	5.458
36.586		32.536	5.5254
36.598		32.808	5.5928
36.61		33.077	5.6601
36.622		33.343	5.7274
36.634		33.607	5.7946
36.646		33.868	5.8618
36.658	1.6868e-010	34.126	5.9289
36.67		34.382	5.996
36.682		34.636	6.0631
36.694		34.888	6.1301
36.706		35.138	6.197
36.718		35.385	6.2639
36.73		35.63	6.3308
36.741		35.873	6.3976
36.753		36.115	6.4644
36.765		36.564	6.5311
36.777		37.035	6.5977
36.789		37.504	6.6644
36.801		37.972	6.7309
36.813		38.437	6.7974
36.825		38.901	6.8638
36.837		39.364	6.9302
36.849		39.824	6.9966
36.861		40.283	7.0628
36.873		40.739	7.1291
36.885		41.194	7.1952
36.897		41.647	7.2613
36.909		42.099	7.3274
36.921	1.6869e-010	42.548	7.3934
36.933		42.995	7.4593
36.945		43.441	7.5251
36.957		43.885	7.5909
36.969		44.326	7.6567
36.981		44.766	7.7224
36.993		45.205	7.788
37.005		45.641	7.8535
37.017		46.075	7.919
37.029		46.508	7.9844
37.041		46.938	8.0498
37.052	1.687e-010	47.338	8.1107
37.062	2.0216e-008	47.679	8.0546
37.072	3.6825e-009	48.024	8.0226
37.086	1.5847e-009	48.487	7.9972

37.101	1.8673e-010	48.983	7.9793
37.117	3.2967e-010	49.466	7.9665
37.133	1.747e-010	49.91	7.9572
37.15	9.8368e-011	50.301	7.9502
37.168	1.606e-011	50.633	7.945
37.187	6.8923e-012	50.906	7.9411
37.206	8.8915e-012	51.119	7.938
37.227	1.3237e-011	51.272	7.9356
37.248	1.362e-011	51.367	7.9336
37.271	1.3695e-011	51.405	7.9319
37.295	1.3709e-011	51.387	7.9303
37.32	1.3711e-011	51.314	7.9288
37.347	1.3712e-011	51.189	7.9271
37.374		51.02	7.9252
37.402		50.813	7.9232
37.43		50.573	7.9209
37.459		50.305	7.9183
37.489		50.015	7.9153
37.518	1.3713e-011	49.705	7.9121
37.548		49.381	7.9086
37.578		49.047	7.9048
37.608		48.704	7.9007
37.638		48.357	7.8964
37.668		48.005	7.8918
37.699		47.651	7.887
37.729		47.295	7.8819
37.76		46.938	7.8767
37.791		46.581	7.8712
37.822	1.3714e-011	46.223	7.8655
37.854	1.3713e-011	45.865	7.8596
37.886	1.3714e-011	45.508	7.8535
37.918		45.152	7.8472
37.95		44.797	7.8408
37.983		44.443	7.8341
38.016		44.091	7.8273
38.035		43.892	7.8234
38.053		43.695	7.8194
38.064	1.5573e-009	43.588	7.8029
38.074	1.576e-009	43.483	7.7902
38.088	6.6803e-010	43.336	7.7741
38.122	1.9597e-010	42.963	7.7385
38.157	3.1691e-011	42.56	7.7044
38.192	1.0489e-011	42.128	7.6706
38.228	8.7356e-012	41.67	7.637
38.264	6.9092e-012	41.191	7.6031
38.301	6.6553e-012	40.695	7.5689
38.338	6.6234e-012	40.186	7.5343
38.376	6.6198e-012	39.666	7.4992
38.415	6.6196e-012	39.137	7.4635
38.454	6.6195e-012	38.602	7.4273
38.494		38.061	7.3904
38.535	6.6196e-012	37.516	7.3529
38.577	6.6195e-012	36.966	7.3147
38.619	6.6198e-012	36.413	7.2757
38.663	6.6199e-012	35.856	7.2359
38.708	6.62e-012	35.296	7.1953
38.754	6.6197e-012	34.734	7.1538
38.801	6.6198e-012	34.574	7.1113
38.849	6.6201e-012	34.929	7.0679
38.899	6.62e-012	35.275	7.0234
38.95	6.6206e-012	35.614	6.9779

39.003	6.6201e-012	35.944	6.9312
39.055		36.248	6.8858
39.065	8.7857e-009	36.307	6.8472
39.075	6.5453e-009	36.364	6.8121
39.09	8.1065e-010	36.45	6.7637
39.125	4.9101e-010	36.641	6.6613
39.16	1.4995e-010	36.824	6.5638
39.195	6.2508e-011	36.999	6.4687
39.231	6.2697e-011	37.167	6.3751
39.266	5.813e-011	37.328	6.2824
39.301	5.7529e-011	37.481	6.1904
39.336	5.7451e-011	37.626	6.0988
39.371	5.7441e-011	37.763	6.0076
39.406	5.744e-011	37.892	5.9167
39.441	5.7441e-011	38.013	5.8261
39.477		38.125	5.7358
39.512	5.7442e-011	38.229	5.6457
39.547		38.323	5.556
39.582		38.408	5.4676
39.617	5.7443e-011	38.483	5.3849
39.652		38.549	5.3078
39.687	5.7444e-011	38.604	5.2382
39.722		38.65	5.174
39.758		38.685	5.1141
39.793	5.7445e-011	38.71	5.0578
39.828		38.725	5.0042
39.863	5.7446e-011	38.73	4.9529
39.898		38.725	4.9045
39.933	5.7447e-011	38.709	4.8579
39.968		38.683	4.8127
40.004		38.648	4.7689
40.039	5.7448e-011	38.603	4.7265
40.056		38.577	4.7059
40.066	9.2898e-010	38.562	4.6802
40.076	3.2017e-010	38.546	4.6584
40.1	6.8329e-010	38.504	4.6147
40.148	1.6125e-010	38.403	4.5397
40.199	2.0764e-011	38.276	4.4687
40.25	3.4459e-011	38.129	4.4028
40.301	3.8866e-011	37.964	4.3416
40.353	3.9198e-011	37.782	4.2842
40.404	3.9231e-011	37.585	4.2301
40.456	3.9234e-011	37.372	4.1788
40.507		37.146	4.13
40.559	3.9235e-011	36.908	4.0834
40.61		36.657	4.0387
40.661	3.9236e-011	36.396	3.9959
40.713		36.125	3.9549
40.764	3.9237e-011	35.845	3.9155
40.816		35.556	3.8777
40.867	3.9238e-011	35.26	3.8413
40.919		34.957	3.8065
40.97		34.647	3.7738
41.021	3.9239e-011	34.331	3.7445
41.058		34.105	3.7267
41.068	3.5416e-010	34.042	3.7142
41.078	3.9648e-010	33.98	3.7037
41.108	3.3231e-010	33.789	3.6773
41.172	4.9212e-011	33.376	3.6294
41.24	1.3585e-011	32.933	3.5832
41.31	2.7191e-011	32.463	3.5384

41.382	2.7973e-011	31.977	3.4954
41.454	2.8029e-011	31.487	3.4549
41.526	2.8033e-011	30.993	3.4163
41.598	2.8034e-011	30.496	3.3794
41.67		29.998	3.344
41.742	2.8035e-011	29.499	3.3102
41.814		29.001	3.278
41.886	2.8036e-011	28.503	3.2476
41.958		28.006	3.2193
42.03	2.8037e-011	27.511	3.1932
42.059		27.309	3.183
42.069	9.4316e-011	27.24	3.1744
42.079	3.9531e-010	27.172	3.1669
42.109	1.7632e-010	26.966	3.1473
42.14	7.7662e-011	26.759	3.1287
42.225	1.1225e-011	26.179	3.0795
42.312	2.0045e-011	25.588	3.0319
42.403	2.0706e-011	24.984	2.9854
42.495	2.0745e-011	24.374	2.94
42.588	2.0748e-011	23.761	2.896
42.683		23.149	2.8533
42.779	2.0749e-011	22.541	2.812
42.875		21.937	2.7721
42.972		21.341	2.7338
43.061	2.075e-011	20.804	2.7002
43.071	4.4826e-011	20.744	2.6928
43.081	2.6657e-010	20.683	2.6863
43.111	9.4409e-011	20.503	2.6687
43.16	2.7627e-011	20.209	2.6415
43.26	7.6574e-012	19.624	2.5894
43.36	1.5552e-011	19.049	2.54
43.46	1.5777e-011	18.485	2.493
43.56	1.5789e-011	17.933	2.4483
43.661	1.579e-011	17.392	2.4059
43.761		16.864	2.3657
43.861		16.347	2.3275
43.961	1.5791e-011	15.843	2.2913
44.062		15.345	2.2567
44.072	9.192e-011	15.296	2.2508
44.082	1.3963e-010	15.247	2.2455
44.112	5.0512e-011	15.1	2.231
44.173	5.47e-012	14.805	2.2034
44.274	7.5049e-012	14.335	2.1611
44.374	1.2153e-011	13.876	2.1213
44.474	1.2283e-011	13.428	2.0837
44.574	1.229e-011	12.99	2.0483
44.674	1.2291e-011	12.563	2.015
44.774		12.147	1.9837
44.874		11.742	1.9543
44.975		11.347	1.9269
45.063		11.004	1.9042
45.073	1.1082e-010	10.965	1.8999
45.083	1.6196e-010	10.927	1.896
45.113	2.7887e-011	10.813	1.8854
45.193	1.5361e-011	10.515	1.8596
45.293	7.2724e-012	10.15	1.8298
45.394	9.6625e-012	9.7945	1.8021
45.494	9.7367e-012	9.4481	1.7762
45.594	9.7408e-012	9.1108	1.752
45.694	9.7414e-012	8.7823	1.7294
45.794	9.7415e-012	8.4626	1.7082

45.894	9.7418e-012	8.1514	1.6884
45.994	9.742e-012	7.8485	1.67
46.065	9.7422e-012	7.6396	1.6577
46.075	1.1305e-010	7.6099	1.6547
46.085	1.8456e-010	7.5974	1.6519
46.115	1.7802e-011	7.5907	1.6444
46.205	1.3005e-011	7.5574	1.6238
46.305	6.3576e-012	7.5099	1.6026
46.405	7.7793e-012	7.458	1.5826
46.505	7.8288e-012	7.4055	1.5639
46.606	7.8317e-012	7.3547	1.5462
46.706	7.8319e-012	7.3067	1.5296
46.806	7.8321e-012	7.2621	1.514
46.906	7.8324e-012	7.2209	1.4993
47.006	7.8325e-012	7.1833	1.4855
47.066	7.8327e-012	7.1621	1.4776
47.076	1.0641e-010	7.1584	1.4753
47.086	1.8624e-010	7.1548	1.4732
47.116	1.4276e-011	7.1438	1.4675
47.206	9.7522e-012	7.1011	1.4517
47.307	5.2869e-012	7.0461	1.4354
47.407	6.3243e-012	6.9881	1.4199
47.507	6.3626e-012	6.9302	1.4053
47.607	6.3646e-012	6.8741	1.3914
47.707	6.3649e-012	6.8204	1.3782
47.807	6.3651e-012	6.7698	1.3657
47.907	6.3652e-012	6.7222	1.3539
48.008	6.3654e-012	6.6778	1.3428
48.068	6.3655e-012	6.6523	1.3363
48.078	9.5813e-011	6.6479	1.3344
48.088	1.7092e-010	6.6436	1.3327
48.118	1.2837e-011	6.6306	1.3281
48.208	7.5074e-012	6.5843	1.3153
48.308	4.4048e-012	6.5271	1.3018
48.408	5.1816e-012	6.4679	1.289
48.508	5.2116e-012	6.4091	1.2767
48.608	5.2132e-012	6.3519	1.2651
48.708	5.2134e-012	6.297	1.254
48.809	5.2136e-012	6.2446	1.2434
48.909	5.2137e-012	6.195	1.2333
49.009	5.2139e-012	6.1481	1.2238
49.069		6.121	1.2183
49.079	8.3085e-011	6.1164	1.2167
49.089	1.433e-010	6.1118	1.2153
49.119	1.1453e-011	6.0981	1.2114
49.209	5.9452e-012	6.0512	1.2006
49.309	3.6643e-012	5.9947	1.1892
49.409	4.2646e-012	5.9367	1.1783
49.51	4.2887e-012	5.8793	1.1679
49.61	4.29e-012	5.8233	1.158
49.71	4.2902e-012	5.7694	1.1486
49.81	4.2903e-012	5.7178	1.1396
49.91	4.2904e-012	5.6685	1.131
50.01	4.2905e-012	5.6215	1.1229
50.07		5.5942	1.1183
50.08	6.9392e-011	5.5896	1.117
50.09	1.1868e-010	5.585	1.1158
50.12	9.6585e-012	5.5712	1.1125
50.211	4.8484e-012	5.5253	1.1034
50.311	3.0326e-012	5.4709	1.0939
50.411	3.5137e-012	5.4154	1.0848

50.511	3.5334e-012	5.3605	1.0761
50.611	3.5345e-012	5.3069	1.0678
50.711	3.5347e-012	5.2551	1.06
50.811		5.2053	1.0524
50.912	3.5348e-012	5.1575	1.0452
51.012		5.1118	1.0382
51.072	3.535e-012	5.0851	1.0341
51.082	5.5412e-011	5.0806	1.0329
51.092	9.6541e-011	5.0761	1.0319
51.122	7.5608e-012	5.0627	1.029
51.212	4.0744e-012	5.0186	1.0211
51.312	2.4851e-012	4.9669	1.0126
51.412	2.8866e-012	4.9144	1.0043
51.512	2.903e-012	4.8624	0.99629
51.613	2.904e-012	4.8117	0.9885
51.713	2.9041e-012	4.7625	0.98093
51.813		4.715	0.97357
51.913	2.9042e-012	4.6694	0.96643
52.013	2.9043e-012	4.6255	0.9595
52.073	2.9044e-012	4.5998	0.95542
52.083	4.138e-011	4.5955	0.95436
52.093	7.3823e-011	4.5912	0.95338
52.123	5.4957e-012	4.5783	0.95067
52.213	3.535e-012	4.5365	0.943
52.314	2.0027e-012	4.4877	0.93476
52.414	2.3526e-012	4.4383	0.92676
52.514	2.3666e-012	4.3894	0.91896
52.614	2.3674e-012	4.3415	0.91137
52.714	2.3675e-012	4.2951	0.90396
52.814		4.2502	0.89674
52.914		4.2069	0.88972
53.015	2.3676e-012	4.1651	0.88287
53.075	2.3677e-012	4.1407	0.87884
53.085	1.7069e-009	4.1375	0.88417
53.095	1.032e-009	4.1341	0.88821
53.125	3.0792e-010	4.1272	0.89728
53.215	4.1018e-011	4.1787	0.91967
53.315	1.4082e-011	4.2996	0.94314
53.415	1.1018e-011	4.4561	0.96623
53.515	1.0799e-011	4.6299	0.98923
53.615	1.0788e-011	4.8107	1.0122
53.716		4.9924	1.0352
53.816		5.1719	1.0582
53.916		5.3475	1.0811
54.016		5.5181	1.1041
54.076		5.6185	1.1178
54.086	1.1916e-012	5.6352	1.1201
54.096	1.3879e-011	5.6519	1.1224
54.126	1.2952e-011	5.7013	1.1293
54.216	1.0953e-011	5.8452	1.1498
54.316	1.0814e-011	5.9999	1.1725
54.417	1.0806e-011	6.1492	1.1952
54.517		6.2935	1.2177
54.617		6.4328	1.2401
54.717	1.0807e-011	6.5675	1.2624
54.817		6.6978	1.2845
54.917		6.8239	1.3066
55.017		6.946	1.3284
55.078		7.0178	1.3415
55.088	1.0846e-009	7.0288	1.3375
55.098	2.3787e-010	7.0401	1.3349

55.128	7.8962e-011	7.0701	1.33
55.218	1.0924e-011	7.0846	1.3205
55.318	3.2889e-012	7.0333	1.3113
55.418	1.8221e-012	6.9436	1.3026
55.518	2.1371e-012	6.8344	1.294
55.618	2.1488e-012	6.7165	1.2854
55.718	2.1494e-012	6.5961	1.2769
55.819	2.1495e-012	6.4766	1.2685
55.919	2.1496e-012	6.3599	1.26
56.019		6.247	1.2517
56.079		6.1808	1.2467
56.089	2.5018e-010	6.1696	1.2444
56.099	7.6022e-011	6.1585	1.2424
56.129	4.4765e-011	6.1247	1.2372
56.219	1.1197e-012	6.0092	1.2228
56.319	2.4816e-013	5.8699	1.2073
56.419	4.0318e-014	5.7822	1.1922
56.52	2.3727e-014	5.7709	1.1773
56.62	2.6893e-014	5.7536	1.1626
56.72	2.707e-014	5.731	1.1481
56.82	2.7073e-014	5.7037	1.1338
56.92	2.706e-014	5.6724	1.1198
57.02	2.7067e-014	5.6377	1.106
57.08	2.7075e-014	5.6159	1.0978
57.09	3.5283e-012	5.6122	1.0964
57.1	1.1583e-012	5.6085	1.0951
57.13	3.8356e-013	5.5971	1.091
57.22	1.8062e-015	5.5612	1.0789
57.321	1.9489e-015	5.5192	1.0657
57.421	6.0121e-016	5.5461	1.0528
57.521	3.2076e-016	5.5748	1.0402
57.621	3.7666e-016	5.5993	1.0278
57.721	4.8055e-016	5.6201	1.0156
57.821	3.5584e-016	5.6374	1.0037
57.921	3.1473e-016	5.6515	0.99206
58.022	3.7574e-016	5.6625	0.98067
58.082	5.6646e-016	5.6681	0.97393
58.092	5.4002e-016	5.669	0.97281
58.102	5.1271e-016	5.6699	0.9717
58.132	1.7738e-016	5.6723	0.96837
58.222	1.7552e-016	5.6773	0.95857
58.322	2.1401e-016	5.6806	0.94791
58.422	8.8156e-017	5.6816	0.9375
58.522	2.2001e-016	5.6806	0.92733
58.622	1.0745e-016	5.6778	0.91739
58.723	1.8871e-016	5.6732	0.90768
58.823	1.9751e-016	5.667	0.8982
58.923	1.7232e-016	5.6594	0.88894
59.023	2.8296e-017	5.6505	0.87991
59.083	1.9932e-016	5.6447	0.87458
59.093	7.8662e-011	5.6437	0.87407
59.103	5.7656e-011	5.6427	0.87348
59.133	1.916e-011	5.6396	0.87155
59.223	8.6287e-013	5.6294	0.86559
59.323	6.3909e-013	5.6171	0.85906
59.424	5.4234e-013	5.604	0.8527
59.524	5.2844e-013	5.5902	0.84653
59.624	5.2771e-013	5.576	0.84055
59.724	5.277e-013	5.5614	0.83474
59.824		5.5465	0.8291
59.924	5.2774e-013	5.5315	0.82363

60.024		5.5164	0.81832
60.084	5.2776e-013	5.5073	0.81521
60.095	1.0863e-011	5.5058	0.81472
60.105	9.0323e-012	5.5043	0.81422
60.135	1.992e-012	5.4997	0.81273
60.225	7.0245e-013	5.4861	0.80833
60.325	5.7556e-013	5.471	0.80356
60.425	5.6026e-013	5.456	0.79893
60.525	5.5946e-013	5.441	0.79443
60.625	5.5942e-013	5.4262	0.79004
60.725		5.4114	0.78578
60.826	5.5946e-013	5.3968	0.78163
60.926		5.3823	0.77757
61.026	5.5945e-013	5.368	0.7736
61.086	5.5947e-013	5.3594	0.77125
61.096	1.3808e-011	5.358	0.77081
61.106	1.5178e-011	5.3566	0.77037
61.136	1.4794e-012	5.3523	0.76912
61.226	4.3231e-013	5.3396	0.76544
61.326	4.4166e-013	5.3257	0.76145
61.426	4.7899e-013	5.3118	0.75754
61.527	4.8107e-013	5.2981	0.7537
61.627	4.8116e-013	5.2844	0.74992
61.727	4.8114e-013	5.2708	0.74621
61.827	4.8119e-013	5.2574	0.74255
61.927	4.8122e-013	5.2439	0.73896
62.027	4.8119e-013	5.2306	0.73543
62.087	4.8121e-013	5.2226	0.73333
62.097	8.5415e-012	5.2212	0.73292
62.107	1.3895e-011	5.2199	0.73253
62.137	1.2773e-012	5.2159	0.73138
62.227	5.133e-013	5.204	0.72803
62.328	3.5777e-013	5.1908	0.72438
62.428	3.968e-013	5.1776	0.72079
62.528	3.9896e-013	5.1645	0.71726
62.628	3.9909e-013	5.1514	0.71377
62.728	3.9911e-013	5.1383	0.71034
62.828	3.9909e-013	5.1253	0.70695
62.928	3.9911e-013	5.1122	0.70361
63.029	3.9914e-013	5.0991	0.70033
63.089	3.9913e-013	5.0913	0.69837
63.099	4.5148e-012	5.09	0.69799
63.109	7.26e-012	5.0887	0.69762
63.139	7.5144e-013	5.0848	0.69655
63.229	5.4243e-013	5.073	0.69341
63.329	2.7439e-013	5.06	0.68999
63.429	3.1737e-013	5.0469	0.68661
63.529	3.1945e-013	5.0338	0.68329
63.629	3.1955e-013	5.0207	0.68001
63.73	3.1959e-013	5.0076	0.67677
63.83		4.9945	0.67357
63.93	3.1958e-013	4.9813	0.67042
64.03		4.9682	0.66731
64.09	3.196e-013	4.9603	0.66549
64.1	1.7356e-012	4.9589	0.66514
64.11	2.8326e-012	4.9576	0.66479
64.14	1.0819e-012	4.9537	0.66378
64.23	4.0469e-013	4.9418	0.66081
64.33	2.0257e-013	4.9285	0.65757
64.431	2.4537e-013	4.9152	0.65438
64.531	2.4725e-013	4.9019	0.65123

64.631	2.4737e-013	4.8885	0.64811
64.731		4.8751	0.64504
64.831	2.4736e-013	4.8617	0.642
64.931	2.4738e-013	4.8482	0.639
65.031	2.4737e-013	4.8347	0.63604
65.092	2.474e-013	4.8265	0.63427
65.102	3.501e-013	4.8252	0.63395
65.112	4.003e-012	4.8238	0.63362
65.142	1.3322e-012	4.8198	0.63266
65.232	2.399e-013	4.8075	0.62985
65.332	1.4576e-013	4.7939	0.62678
65.432	1.8339e-013	4.7802	0.62374
65.532	1.8501e-013	4.7665	0.62074
65.632	1.8511e-013	4.7528	0.61778
65.732	1.8513e-013	4.739	0.61485
65.833	1.8512e-013	4.7251	0.61195
65.933	1.8514e-013	4.7113	0.60909
66.033		4.6974	0.60626
66.093	1.8512e-013	4.689	0.60457
66.103	1.1066e-012	4.6876	0.60424
66.113	2.3172e-012	4.6862	0.60393
66.143	1.3711e-012	4.682	0.60302
66.233	1.0327e-013	4.6695	0.60036
66.333	1.0285e-013	4.6554	0.59744
66.433	1.325e-013	4.6414	0.59456
66.534	1.3388e-013	4.6272	0.59171
66.634	1.3395e-013	4.6131	0.58889
66.734	1.3396e-013	4.5989	0.58611
66.834		4.5847	0.58334
66.934	1.3397e-013	4.5705	0.58061
67.034	1.3396e-013	4.5562	0.57791
67.094	1.3398e-013	4.5477	0.5763
67.104	1.496e-012	4.5463	0.57598
67.114	1.2168e-012	4.5448	0.57569
67.144	1.2625e-012	4.5406	0.57483
67.235	3.5009e-014	4.5277	0.57231
67.335	7.179e-014	4.5133	0.56955
67.435	9.2501e-014	4.4989	0.56682
67.535	9.3558e-014	4.4845	0.56411
67.635	9.3639e-014	4.4701	0.56144
67.735	9.3649e-014	4.4556	0.55878
67.835	9.3674e-014	4.4411	0.55616
67.936	9.366e-014	4.4267	0.55356
68.036	9.3654e-014	4.4122	0.55098
68.096	9.3679e-014	4.4035	0.54945
68.106	1.5254e-012	4.402	0.54916
68.116	5.6566e-013	4.4006	0.54888
68.146	7.9126e-013	4.3962	0.54807
68.236	3.8478e-014	4.3831	0.5457
68.336	4.7384e-014	4.3686	0.54309
68.436	6.2295e-014	4.354	0.54051
68.536	6.3128e-014	4.3394	0.53795
68.637	6.317e-014	4.3248	0.53541
68.737	6.3162e-014	4.3102	0.5329
68.837	6.3185e-014	4.2956	0.53041
68.937	6.3189e-014	4.281	0.52794
69.037	6.3174e-014	4.2664	0.5255
69.097	6.3185e-014	4.2576	0.52404
69.107	1.3541e-012	4.2562	0.52378
69.117	2.4879e-013	4.2547	0.52353
69.147	7.4596e-013	4.2503	0.52277

69.237	4.8091e-014	4.2372	0.52053
69.338	2.9229e-014	4.2225	0.51808
69.438	4.0371e-014	4.2079	0.51564
69.538	4.1019e-014	4.1932	0.51323
69.638	4.103e-014	4.1786	0.51083
69.738	4.1033e-014	4.164	0.50846
69.838	4.1056e-014	4.1494	0.50611
69.938	4.1062e-014	4.1348	0.50377
70.039	4.104e-014	4.1203	0.50146
70.099	4.1057e-014	4.1115	0.50008
70.109	1.0981e-012	4.1101	0.49984
70.119	1.7548e-013	4.1086	0.49961
70.149	5.2457e-013	4.1042	0.4989
70.239	4.6513e-014	4.0911	0.4968
70.339	1.7008e-014	4.0765	0.49449
70.439	2.5186e-014	4.062	0.4922
70.539	2.5622e-014	4.0474	0.48993
70.639	2.5641e-014	4.0329	0.48768
70.739	2.5646e-014	4.0184	0.48544
70.84	2.5652e-014	4.0039	0.48322
70.94	2.5641e-014	3.9895	0.48102
71.04	2.5657e-014	3.9751	0.47884
71.1	2.5656e-014	3.9665	0.47754

Material Data

Aluminum Alloy

TABLE 25
Aluminum Alloy > Constants

Density	2.77e-006 kg mm^-3
Coefficient of Thermal Expansion	2.3e-005 K^-1
Specific Heat	8.75e+005 mJ kg^-1 K^-1

TABLE 26
Aluminum Alloy > Color

Red	Green	Blue
138	104	46

TABLE 27
Aluminum Alloy > Compressive Ultimate Strength

Compressive Ultimate Strength MPa
0

TABLE 28
Aluminum Alloy > Compressive Yield Strength

Compressive Yield Strength MPa
280

TABLE 29
Aluminum Alloy > Tensile Yield Strength

Tensile Yield Strength MPa
280

TABLE 30
Aluminum Alloy > Tensile Ultimate Strength

Tensile Ultimate Strength MPa
310

TABLE 31
Aluminum Alloy > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature K
295.15

TABLE 32
Aluminum Alloy > Isotropic Thermal Conductivity

Thermal Conductivity W mm^-1 K^-1	Temperature K
0.114	173.15
0.144	273.15
0.165	373.15
0.175	473.15

TABLE 33
Aluminum Alloy > S-N Curve

Alternating Stress MPa	Cycles	R-Ratio
275.8	1700	-1
241.3	5000	-1
206.8	34000	-1
172.4	1.4e+005	-1
137.9	8.e+005	-1
117.2	2.4e+006	-1
89.63	5.5e+007	-1
82.74	1.e+008	-1
170.6	50000	-0.5
139.6	3.5e+005	-0.5
108.6	3.7e+006	-0.5
87.91	1.4e+007	-0.5
77.57	5.e+007	-0.5
72.39	1.e+008	-0.5
144.8	50000	0
120.7	1.9e+005	0
103.4	1.3e+006	0
93.08	4.4e+006	0
86.18	1.2e+007	0
72.39	1.e+008	0
74.12	3.e+005	0.5
70.67	1.5e+006	0.5
66.36	1.2e+007	0.5
62.05	1.e+008	0.5

TABLE 34
Aluminum Alloy > Isotropic Resistivity

Resistivity ohm mm	Temperature K
2.43e-005	273.15
2.67e-005	293.15
3.63e-005	373.15

TABLE 35
Aluminum Alloy > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
71000	0.33	69608	26692	

TABLE 36
Aluminum Alloy > Isotropic Relative Permeability

Relative Permeability
1

TBC coating

TABLE 37
TBC coating > Constants

Density	4.85e-006 kg mm^-3
Specific Heat	4.8e+005 mJ kg^-1 K^-1
Thermal Conductivity	2.e-003 W mm^-1 K^-1

TABLE 38
TBC coating > Color

Red	Green	Blue
155	244	255

TABLE 39
TBC coating > Melting Temperature

Melting Temperature K
2773.2

TABLE 40
TBC coating > Isotropic Elasticity

Young's Modulus MPa	Poisson's Ratio	Bulk Modulus MPa	Shear Modulus MPa	Temperature K
1.75e+005	0.275	1.2963e+005	68627	